Validation Report

Traceability of the FCA Formal Context to the EO Discourse

Source: Dietz & Mulder, Enterprise Ontology, 2nd ed., Springer, 2024
Framework: Formal Concept Analysis (Ganter & Wille) | Domain: Thing

Foundational Reasoning Principles

The following principles, grounded exclusively in the EO discourse and FCA framework, govern all assessments in this report.

Concept primacy over symbol (Constraint 1). References are assessed at the concept level. A formal context entry is traceable if the cited discourse sentence expresses or implies the concept, even when different symbols, synonyms, or bracket conventions are used.

Composite classification. The formal context enforces a disjoint & complete Composite / Elemental partition. No non-placeholder item is asserted Elemental anywhere in the discourse. Therefore every non-placeholder item whose discourse context implies component parts — biological organisms (56-19, 66-25), physical structures (66-05, 66-26), contracts and legal instruments (59-08), conceptual mirrors of composite things (56-28) — is classified Composite. The burden of Elemental classification requires a positive discourse assertion of atomic indivisibility; the discourse provides none for these items.

Noetic thing as placeholder superconcept (Constraint 3). 'Noetic thing' does not appear anywhere in the source text. It is a placeholder term introduced by the researchers, transparent per Constraint 3. It is structurally required as the superconcept of Conceptual thing and Abstract thing: both are 'in the mind' (59-10; 54-01 + 58-04), and they differ solely in the representation relation — Conceptual things represent Concrete things (56-21); Abstract things do not (58-03). Placeholder characteristics and items carry no discourse reference by design and are not penalised.

Abstract things are in the mind. 54-01: thoughts are things in the mind. 58-04: Abstract things are conceived in the mind through reasoning by analogy. 58-07: you can count 387 dogs but cannot observe 387 itself — the number 387 exists as a thought in the mind, not in reality. Both Conceptual things (59-10) and Abstract things therefore share 'in a mind'; their distinction is solely the representation relation.

Abstract things are representable by Conceptual things. The discourse uses (the number '387') — parenthesis notation indicating a Conceptual thing (Constraint 8 bracket convention). This Conceptual thing in the mind represents the Abstract thing <387>. Sentence 56-21 gives the primary case of representation (Concrete things) but never states it is the only case. Sentence 58-03 says Abstract things do not represent Concrete things — this concerns Abstract things as the representing party, not whether Abstract things can be represented.

1. Summary Table

Type	Entry	Ref	Outcome
Item (C1)	56-05 [a cloud]	56-05	PASS
Item (C1)	56-12 [a body]	56-12	PASS
Item (C1)	56-19 [a human]	56-19	PASS
Item (C1)	56-19 [a dog]	56-19	PASS
Item (C1)	56-19 [a membership]	56-19	PASS
Item (C1)	56-19 [a rental]	56-19	PASS
Item (C1)	56-20 <a number>	56-20	PASS
Item (C1)	56-20 <a boolean value>	56-20	PASS
Item (C1)	56-33 [the human 'Charlie Chaplin']	56-33	PASS
Item (C1)	56-33 (the human 'Charlie Chaplin')	56-33	PASS
Item (C1)	59-04 <a date>	59-04	PASS

Type	Entry	Ref	Outcome
Item (C1)	59-08 [an insurance policy]	59-08	PASS
Item (C1)	59-08 [an employment contract]	59-08	PASS
Item (C1)	59-08 [a car]	59-08	PASS
Item (C1)	59-08 [a house]	59-08	PASS
Item (C1)	62-21 [a student]	62-21	PASS
Item (C1)	65-02 (a chair)	65-02	PASS
Item (C1)	66-05 [a ship]	66-05	PASS
Item (C1)	66-05 [a plank]	66-05	PASS
Item (C1)	66-14 [a butterfly]	66-14	PASS
Item (C1)	66-14 [a caterpillar]	66-14	PASS
Item (C1)	66-25 [a brain]	66-25	PASS
Item (C1)	66-26 [a motor block]	66-26	PASS
Char (C2)	54-01 in a mind	54-01	PASS
Char (C2)	54-02 can represent a Thing	54-02	PASS
Char (C2)	54-02 in reality	54-02	PASS
Char (C2)	54-02 can be represented by a Conceptual thing	54-02	PASS
Char (C2)	55-12 collection of Things	55-12	PASS
Char (C2)	56-01 distinguishable	56-01	PASS
Char (C2)	56-02 with Object	56-02	PASS
Char (C2)	56-02 with assigned Features	56-02	PASS
Char (C2)	56-29 physical dimensions	56-29	PASS
Char (C2)	56-29 conceived	56-29	PASS
Char (C2)	58-03 cannot represent a Thing	58-03	PASS
Char (C2)	56-19 Concrete thing	56-19	PASS
Char (C2)	56-20 Abstract thing	56-20	PASS
Char (C2)	56-29 Tangible concrete thing	56-29	PASS
Char (C2)	56-29 Intangible concrete thing	56-29	PASS
Char (C2)	56-33 Conceptual thing	56-33	PASS
Char (C2)	59-01 Thing	59-01	PASS
Char (C2)	66-05 Composite thing	66-05	PASS
Rel key (C3)	[a cloud] — with Object	56-05	PASS
Rel key (C3)	[a cloud] — with assigned Features	56-05	PASS
Rel key (C3)	[a cloud] — Thing	56-05	PASS
Rel key (C3)	[a body] — collection of Things	66-25	PASS
Rel key (C3)	[a body] — with Object	56-12	PASS
Rel key (C3)	[a body] — with assigned Features	56-12	PASS
Rel key (C3)	[a body] — Composite thing	66-25	PASS

Type	Entry	Ref	Outcome
Rel key (C3)	[a human] — in reality	59-08	PASS
Rel key (C3)	[a human] — physical dimensions	56-29	PASS
Rel key (C3)	[a human] — Concrete thing	56-19	PASS
Rel key (C3)	[a human] — Tangible concrete thing	56-29	PASS
Rel key (C3)	[a human] — Thing	59-08	PASS
Rel key (C3)	[a human] — Comcposite concrete thing	56-19	PASS
Rel key (C3)	[a dog] — physical dimensions	56-29	PASS
Rel key (C3)	[a dog] — Concrete thing	56-19	PASS
Rel key (C3)	[a dog] — Tangible concrete thing	56-29	PASS
Rel key (C3)	[a dog] — Comcposite concrete thing	56-19	PASS
Rel key (C3)	[a membership] — conceived	56-29	PASS
Rel key (C3)	[a membership] — Concrete thing	56-19	PASS
Rel key (C3)	[a membership] — Intangible concrete thing	56-29	PASS
Rel key (C3)	[a membership] — Comcposite concrete thing	56-19	PASS
Rel key (C3)	[a rental] — in reality	59-08	PASS
Rel key (C3)	[a rental] — conceived	56-29	PASS
Rel key (C3)	[a rental] — Concrete thing	56-19	PASS
Rel key (C3)	[a rental] — Intangible concrete thing	56-29	PASS
Rel key (C3)	[a rental] — Thing	59-08	PASS
Rel key (C3)	[a rental] — Comcposite concrete thing	56-19	PASS
Rel key (C3)	<a number> — Abstract thing	56-20	PASS
Rel key (C3)	<a boolean value> — Abstract thing	56-20	PASS
Rel key (C3)	[the human 'Charlie Chaplin'] — Concrete thing	56-33	PASS
Rel key (C3)	[the human 'Charlie Chaplin'] — Comcposite concrete thing	56-33	PASS
Rel key (C3)	(the human 'Charlie Chaplin') — can represent a Thing	56-21	PASS
Rel key (C3)	(the human 'Charlie Chaplin') — Conceptual thing	56-33	PASS
Rel key (C3)	(the human 'Charlie Chaplin') — Thing	65-02	PASS
Rel key (C3)	<a date> — Abstract thing	59-04	PASS
Rel key (C3)	<a date> — Thing	59-04	PASS
Rel key (C3)	[an insurance policy] — in reality	59-08	PASS
Rel key (C3)	[an insurance policy] — Concrete thing	59-08	PASS
Rel key (C3)	[an insurance policy] — Thing	59-08	PASS
Rel key (C3)	[an insurance policy] — Comcposite concrete thing	59-08	PASS
Rel key (C3)	[an employment contract] — in reality	59-08	PASS
Rel key (C3)	[an employment contract] — Concrete thing	59-08	PASS
Rel key (C3)	[an employment contract] — Thing	59-08	PASS
Rel key (C3)	[an employment contract] — Comcposite concrete thing	59-08	PASS

Type	Entry	Ref	Outcome
Rel key (C3)	[a car] — in reality	59-08	PASS
Rel key (C3)	[a car] — collection of Things	66-26	PASS
Rel key (C3)	[a car] — Concrete thing	59-08	PASS
Rel key (C3)	[a car] — Thing	59-08	PASS
Rel key (C3)	[a car] — Composite thing	66-26	PASS
Rel key (C3)	[a car] — Comcposite concrete thing	59-08	PASS
Rel key (C3)	[a house] — in reality	59-08	PASS
Rel key (C3)	[a house] — Concrete thing	59-08	PASS
Rel key (C3)	[a house] — Tangible concrete thing	59-09	PASS
Rel key (C3)	[a house] — Thing	59-08	PASS
Rel key (C3)	[a house] — Comcposite concrete thing	59-08	PASS
Rel key (C3)	[a student] — Concrete thing	62-21	PASS
Rel key (C3)	[a student] — Comcposite concrete thing	62-21	PASS
Rel key (C3)	(a chair) — Thing	65-02	PASS
Rel key (C3)	[a ship] — collection of Things	66-05	PASS
Rel key (C3)	[a ship] — with Object	66-09	PASS
Rel key (C3)	[a ship] — with assigned Features	66-09	PASS
Rel key (C3)	[a ship] — Concrete thing	66-19	PASS
Rel key (C3)	[a ship] — Thing	66-09	PASS
Rel key (C3)	[a ship] — Composite thing	66-05	PASS
Rel key (C3)	[a ship] — Comcposite concrete thing	66-19	PASS
Rel key (C3)	[a plank] — Thing	66-05	PASS
Rel key (C3)	[a butterfly] — with Object	66-14	PASS
Rel key (C3)	[a butterfly] — Thing	66-14	PASS
Rel key (C3)	[a caterpillar] — with Object	66-14	PASS
Rel key (C3)	[a caterpillar] — Thing	66-14	PASS
Rel key (C3)	[a brain] — Thing	66-25	PASS
Rel key (C3)	[a motor block] — Thing	66-26	PASS
Rel - # (C4)	[a cloud] — in reality	-	PASS
Rel - # (C4)	[a cloud] — can be represented by a Conceptual thing	-	PASS
Rel - # (C4)	[a cloud] — collection of Things	-	PASS
Rel - # (C4)	[a cloud] — distinguishable	-	PASS
Rel - # (C4)	[a cloud] — physical dimensions	-	PASS
Rel - # (C4)	[a cloud] — Concrete thing	-	PASS
Rel - # (C4)	[a cloud] — Tangible concrete thing	-	PASS
Rel - # (C4)	[a cloud] — Composite thing	-	PASS
Rel - # (C4)	[a cloud] — Comcposite concrete thing	-	PASS

Type	Entry	Ref	Outcome
Rel - # (C4)	[a cloud] — Composite tangible concrete thing	-	PASS
Rel - # (C4)	[a body] — in reality	-	PASS
Rel - # (C4)	[a body] — can be represented by a Conceptual thing	-	PASS
Rel - # (C4)	[a body] — distinguishable	-	PASS
Rel - # (C4)	[a body] — physical dimensions	-	PASS
Rel - # (C4)	[a body] — Concrete thing	-	PASS
Rel - # (C4)	[a body] — Tangible concrete thing	-	PASS
Rel - # (C4)	[a body] — Thing	-	PASS
Rel - # (C4)	[a body] — Comcposite concrete thing	-	PASS
Rel - # (C4)	[a body] — Composite tangible concrete thing	-	PASS
Rel - # (C4)	[a human] — can be represented by a Conceptual thing	-	PASS
Rel - # (C4)	[a human] — collection of Things	-	PASS
Rel - # (C4)	[a human] — distinguishable	-	PASS
Rel - # (C4)	[a human] — with Object	-	PASS
Rel - # (C4)	[a human] — with assigned Features	-	PASS
Rel - # (C4)	[a human] — Composite thing	-	PASS
Rel - # (C4)	[a human] — Composite tangible concrete thing	-	PASS
Rel - # (C4)	[a dog] — in reality	-	PASS
Rel - # (C4)	[a dog] — can be represented by a Conceptual thing	-	PASS
Rel - # (C4)	[a dog] — collection of Things	-	PASS
Rel - # (C4)	[a dog] — distinguishable	-	PASS
Rel - # (C4)	[a dog] — with Object	-	PASS
Rel - # (C4)	[a dog] — with assigned Features	-	PASS
Rel - # (C4)	[a dog] — Thing	-	PASS
Rel - # (C4)	[a dog] — Composite thing	-	PASS
Rel - # (C4)	[a dog] — Composite tangible concrete thing	-	PASS
Rel - # (C4)	[a membership] — in reality	-	PASS
Rel - # (C4)	[a membership] — can be represented by a Conceptual thing	-	PASS
Rel - # (C4)	[a membership] — collection of Things	-	PASS
Rel - # (C4)	[a membership] — distinguishable	-	PASS
Rel - # (C4)	[a membership] — with Object	-	PASS
Rel - # (C4)	[a membership] — with assigned Features	-	PASS
Rel - # (C4)	[a membership] — Thing	-	PASS
Rel - # (C4)	[a membership] — Composite thing	-	PASS
Rel - # (C4)	[a membership] — Composite intangible concrete thing	-	PASS
Rel - # (C4)	[a rental] — can be represented by a Conceptual thing	-	PASS
Rel - # (C4)	[a rental] — collection of Things	-	PASS

Type	Entry	Ref	Outcome
Rel - # (C4)	[a rental] — distinguishable	-	PASS
Rel - # (C4)	[a rental] — with Object	-	PASS
Rel - # (C4)	[a rental] — with assigned Features	-	PASS
Rel - # (C4)	[a rental] — Composite thing	-	PASS
Rel - # (C4)	[a rental] — Composite intangible concrete thing	-	PASS
Rel - # (C4)	<a number> — in a mind	-	PASS
Rel - # (C4)	<a number> — can be represented by a Conceptual thing	-	PASS
Rel - # (C4)	<a number> — single Thing	-	PASS
Rel - # (C4)	<a number> — distinguishable	-	PASS
Rel - # (C4)	<a number> — with Object	-	PASS
Rel - # (C4)	<a number> — with assigned Features	-	PASS
Rel - # (C4)	<a number> — cannot represent a Thing	-	PASS
Rel - # (C4)	<a number> — Thing	-	PASS
Rel - # (C4)	<a number> — Elemental thing	-	PASS
Rel - # (C4)	<a number> — Noetic thing	-	PASS
Rel - # (C4)	<a number> — Elemental abstract thing	-	PASS
Rel - # (C4)	<a number> — Elemental noetic thing	-	PASS
Rel - # (C4)	<a boolean value> — in a mind	-	PASS
Rel - # (C4)	<a boolean value> — can be represented by a Conceptual thing	-	PASS
Rel - # (C4)	<a boolean value> — single Thing	-	PASS
Rel - # (C4)	<a boolean value> — distinguishable	-	PASS
Rel - # (C4)	<a boolean value> — with Object	-	PASS
Rel - # (C4)	<a boolean value> — with assigned Features	-	PASS
Rel - # (C4)	<a boolean value> — cannot represent a Thing	-	PASS
Rel - # (C4)	<a boolean value> — Thing	-	PASS
Rel - # (C4)	<a boolean value> — Elemental thing	-	PASS
Rel - # (C4)	<a boolean value> — Noetic thing	-	PASS
Rel - # (C4)	<a boolean value> — Elemental abstract thing	-	PASS
Rel - # (C4)	<a boolean value> — Elemental noetic thing	-	PASS
Rel - # (C4)	[the human 'Charlie Chaplin'] — in reality	-	PASS
Rel - # (C4)	[the human 'Charlie Chaplin'] — can be represented by a Conceptual thing	-	PASS
Rel - # (C4)	[the human 'Charlie Chaplin'] — collection of Things	-	PASS
Rel - # (C4)	[the human 'Charlie Chaplin'] — distinguishable	-	PASS
Rel - # (C4)	[the human 'Charlie Chaplin'] — with Object	-	PASS
Rel - # (C4)	[the human 'Charlie Chaplin'] — with assigned Features	-	PASS
Rel - # (C4)	[the human 'Charlie Chaplin'] — physical dimensions	-	PASS

Type	Entry	Ref	Outcome
Rel - # (C4)	[the human 'Charlie Chaplin'] — Tangible concrete thing	-	PASS
Rel - # (C4)	[the human 'Charlie Chaplin'] — Thing	-	PASS
Rel - # (C4)	[the human 'Charlie Chaplin'] — Composite thing	-	PASS
Rel - # (C4)	[the human 'Charlie Chaplin'] — Composite tangible concrete thing	-	PASS
Rel - # (C4)	(the human 'Charlie Chaplin') — in a mind	-	PASS
Rel - # (C4)	(the human 'Charlie Chaplin') — can be represented by a Conceptual thing	-	PASS
Rel - # (C4)	(the human 'Charlie Chaplin') — collection of Things	-	PASS
Rel - # (C4)	(the human 'Charlie Chaplin') — distinguishable	-	PASS
Rel - # (C4)	(the human 'Charlie Chaplin') — with Object	-	PASS
Rel - # (C4)	(the human 'Charlie Chaplin') — with assigned Features	-	PASS
Rel - # (C4)	(the human 'Charlie Chaplin') — Composite thing	-	PASS
Rel - # (C4)	(the human 'Charlie Chaplin') — Noetic thing	-	PASS
Rel - # (C4)	(the human 'Charlie Chaplin') — Composite conceptual thing	-	PASS
Rel - # (C4)	(the human 'Charlie Chaplin') — Composite noetic thing	-	PASS
Rel - # (C4)	<a date> — in a mind	-	PASS
Rel - # (C4)	<a date> — can be represented by a Conceptual thing	-	PASS
Rel - # (C4)	<a date> — single Thing	-	PASS
Rel - # (C4)	<a date> — distinguishable	-	PASS
Rel - # (C4)	<a date> — with Object	-	PASS
Rel - # (C4)	<a date> — with assigned Features	-	PASS
Rel - # (C4)	<a date> — cannot represent a Thing	-	PASS
Rel - # (C4)	<a date> — Elemental thing	-	PASS
Rel - # (C4)	<a date> — Noetic thing	-	PASS
Rel - # (C4)	<a date> — Elemental abstract thing	-	PASS
Rel - # (C4)	<a date> — Elemental noetic thing	-	PASS
Rel - # (C4)	[an insurance policy] — can be represented by a Conceptual thing	-	PASS
Rel - # (C4)	[an insurance policy] — collection of Things	-	PASS
Rel - # (C4)	[an insurance policy] — distinguishable	-	PASS
Rel - # (C4)	[an insurance policy] — with Object	-	PASS
Rel - # (C4)	[an insurance policy] — with assigned Features	-	PASS
Rel - # (C4)	[an insurance policy] — conceived	-	PASS
Rel - # (C4)	[an insurance policy] — Intangible concrete thing	-	PASS
Rel - # (C4)	[an insurance policy] — Composite thing	-	PASS
Rel - # (C4)	[an insurance policy] — Composite intangible concrete thing	-	PASS
Rel - # (C4)	[an employment contract] — can be represented by a Conceptual thing	-	PASS

Type	Entry	Ref	Outcome
Rel - # (C4)	[an employment contract] — collection of Things	-	PASS
Rel - # (C4)	[an employment contract] — distinguishable	-	PASS
Rel - # (C4)	[an employment contract] — with Object	-	PASS
Rel - # (C4)	[an employment contract] — with assigned Features	-	PASS
Rel - # (C4)	[an employment contract] — conceived	-	PASS
Rel - # (C4)	[an employment contract] — Intangible concrete thing	-	PASS
Rel - # (C4)	[an employment contract] — Composite thing	-	PASS
Rel - # (C4)	[an employment contract] — Composite intangible concrete thing	-	PASS
Rel - # (C4)	[a car] — can be represented by a Conceptual thing	-	PASS
Rel - # (C4)	[a car] — distinguishable	-	PASS
Rel - # (C4)	[a car] — with Object	-	PASS
Rel - # (C4)	[a car] — with assigned Features	-	PASS
Rel - # (C4)	[a car] — physical dimensions	-	PASS
Rel - # (C4)	[a car] — Tangible concrete thing	-	PASS
Rel - # (C4)	[a car] — Composite tangible concrete thing	-	PASS
Rel - # (C4)	[a house] — can be represented by a Conceptual thing	-	PASS
Rel - # (C4)	[a house] — collection of Things	-	PASS
Rel - # (C4)	[a house] — distinguishable	-	PASS
Rel - # (C4)	[a house] — with Object	-	PASS
Rel - # (C4)	[a house] — with assigned Features	-	PASS
Rel - # (C4)	[a house] — physical dimensions	-	PASS
Rel - # (C4)	[a house] — Composite thing	-	PASS
Rel - # (C4)	[a house] — Composite tangible concrete thing	-	PASS
Rel - # (C4)	[a student] — in reality	-	PASS
Rel - # (C4)	[a student] — can be represented by a Conceptual thing	-	PASS
Rel - # (C4)	[a student] — collection of Things	-	PASS
Rel - # (C4)	[a student] — distinguishable	-	PASS
Rel - # (C4)	[a student] — with Object	-	PASS
Rel - # (C4)	[a student] — with assigned Features	-	PASS
Rel - # (C4)	[a student] — physical dimensions	-	PASS
Rel - # (C4)	[a student] — Tangible concrete thing	-	PASS
Rel - # (C4)	[a student] — Thing	-	PASS
Rel - # (C4)	[a student] — Composite thing	-	PASS
Rel - # (C4)	[a student] — Composite tangible concrete thing	-	PASS
Rel - # (C4)	(a chair) — in a mind	-	PASS
Rel - # (C4)	(a chair) — can represent a Thing	-	PASS
Rel - # (C4)	(a chair) — can be represented by a Conceptual thing	-	PASS

Type	Entry	Ref	Outcome
Rel - # (C4)	(a chair) — collection of Things	-	PASS
Rel - # (C4)	(a chair) — distinguishable	-	PASS
Rel - # (C4)	(a chair) — with Object	-	PASS
Rel - # (C4)	(a chair) — with assigned Features	-	PASS
Rel - # (C4)	(a chair) — Conceptual thing	-	PASS
Rel - # (C4)	(a chair) — Composite thing	-	PASS
Rel - # (C4)	(a chair) — Noetic thing	-	PASS
Rel - # (C4)	(a chair) — Composite conceptual thing	-	PASS
Rel - # (C4)	(a chair) — Composite noetic thing	-	PASS
Rel - # (C4)	[a ship] — in reality	-	PASS
Rel - # (C4)	[a ship] — can be represented by a Conceptual thing	-	PASS
Rel - # (C4)	[a ship] — distinguishable	-	PASS
Rel - # (C4)	[a ship] — physical dimensions	-	PASS
Rel - # (C4)	[a ship] — Tangible concrete thing	-	PASS
Rel - # (C4)	[a ship] — Composite tangible concrete thing	-	PASS
Rel - # (C4)	[a plank] — in reality	-	PASS
Rel - # (C4)	[a plank] — can be represented by a Conceptual thing	-	PASS
Rel - # (C4)	[a plank] — collection of Things	-	PASS
Rel - # (C4)	[a plank] — distinguishable	-	PASS
Rel - # (C4)	[a plank] — with Object	-	PASS
Rel - # (C4)	[a plank] — with assigned Features	-	PASS
Rel - # (C4)	[a plank] — physical dimensions	-	PASS
Rel - # (C4)	[a plank] — Concrete thing	-	PASS
Rel - # (C4)	[a plank] — Tangible concrete thing	-	PASS
Rel - # (C4)	[a plank] — Composite thing	-	PASS
Rel - # (C4)	[a plank] — Comcposite concrete thing	-	PASS
Rel - # (C4)	[a plank] — Composite tangible concrete thing	-	PASS
Rel - # (C4)	[a butterfly] — in reality	-	PASS
Rel - # (C4)	[a butterfly] — can be represented by a Conceptual thing	-	PASS
Rel - # (C4)	[a butterfly] — collection of Things	-	PASS
Rel - # (C4)	[a butterfly] — distinguishable	-	PASS
Rel - # (C4)	[a butterfly] — with assigned Features	-	PASS
Rel - # (C4)	[a butterfly] — physical dimensions	-	PASS
Rel - # (C4)	[a butterfly] — Concrete thing	-	PASS
Rel - # (C4)	[a butterfly] — Tangible concrete thing	-	PASS
Rel - # (C4)	[a butterfly] — Composite thing	-	PASS
Rel - # (C4)	[a butterfly] — Comcposite concrete thing	-	PASS

Type	Entry	Ref	Outcome
Rel - # (C4)	[a butterfly] — Composite tangible concrete thing	-	PASS
Rel - # (C4)	[a caterpillar] — in reality	-	PASS
Rel - # (C4)	[a caterpillar] — can be represented by a Conceptual thing	-	PASS
Rel - # (C4)	[a caterpillar] — collection of Things	-	PASS
Rel - # (C4)	[a caterpillar] — distinguishable	-	PASS
Rel - # (C4)	[a caterpillar] — with assigned Features	-	PASS
Rel - # (C4)	[a caterpillar] — physical dimensions	-	PASS
Rel - # (C4)	[a caterpillar] — Concrete thing	-	PASS
Rel - # (C4)	[a caterpillar] — Tangible concrete thing	-	PASS
Rel - # (C4)	[a caterpillar] — Composite thing	-	PASS
Rel - # (C4)	[a caterpillar] — Comcposite concrete thing	-	PASS
Rel - # (C4)	[a caterpillar] — Composite tangible concrete thing	-	PASS
Rel - # (C4)	[a brain] — in reality	-	PASS
Rel - # (C4)	[a brain] — can be represented by a Conceptual thing	-	PASS
Rel - # (C4)	[a brain] — collection of Things	-	PASS
Rel - # (C4)	[a brain] — distinguishable	-	PASS
Rel - # (C4)	[a brain] — with Object	-	PASS
Rel - # (C4)	[a brain] — with assigned Features	-	PASS
Rel - # (C4)	[a brain] — physical dimensions	-	PASS
Rel - # (C4)	[a brain] — Concrete thing	-	PASS
Rel - # (C4)	[a brain] — Tangible concrete thing	-	PASS
Rel - # (C4)	[a brain] — Composite thing	-	PASS
Rel - # (C4)	[a brain] — Comcposite concrete thing	-	PASS
Rel - # (C4)	[a brain] — Composite tangible concrete thing	-	PASS
Rel - # (C4)	[a motor block] — in reality	-	PASS
Rel - # (C4)	[a motor block] — can be represented by a Conceptual thing	-	PASS
Rel - # (C4)	[a motor block] — collection of Things	-	PASS
Rel - # (C4)	[a motor block] — distinguishable	-	PASS
Rel - # (C4)	[a motor block] — with Object	-	PASS
Rel - # (C4)	[a motor block] — with assigned Features	-	PASS
Rel - # (C4)	[a motor block] — physical dimensions	-	PASS
Rel - # (C4)	[a motor block] — Concrete thing	-	PASS
Rel - # (C4)	[a motor block] — Tangible concrete thing	-	PASS
Rel - # (C4)	[a motor block] — Composite thing	-	PASS
Rel - # (C4)	[a motor block] — Comcposite concrete thing	-	PASS
Rel - # (C4)	[a motor block] — Composite tangible concrete thing	-	PASS
Absent (C5)	[a cloud] — in a mind	-	PASS

Type	Entry	Ref	Outcome
Absent (C5)	[a cloud] — can represent a Thing	-	PASS
Absent (C5)	[a cloud] — single Thing	-	PASS
Absent (C5)	[a cloud] — cannot represent a Thing	-	PASS
Absent (C5)	[a cloud] — Abstract thing	-	PASS
Absent (C5)	[a cloud] — conceived	-	PASS
Absent (C5)	[a cloud] — Intangible concrete thing	-	PASS
Absent (C5)	[a cloud] — Conceptual thing	-	PASS
Absent (C5)	[a human] — in a mind	-	PASS
Absent (C5)	[a human] — cannot represent a Thing	-	PASS
Absent (C5)	[a human] — Abstract thing	-	PASS
Absent (C5)	[a human] — conceived	-	PASS
Absent (C5)	[a human] — Conceptual thing	-	PASS
Absent (C5)	<a number> — can represent a Thing	-	PASS
Absent (C5)	<a number> — in reality	-	PASS
Absent (C5)	<a number> — Concrete thing	-	PASS
Absent (C5)	<a number> — Conceptual thing	-	PASS
Absent (C5)	(the human 'Charlie Chaplin') — in reality	-	PASS
Absent (C5)	(the human 'Charlie Chaplin') — Concrete thing	-	PASS
Absent (C5)	(the human 'Charlie Chaplin') — cannot represent a Thing	-	PASS
Interp (C6)	54-01: Some thoughts are Conceptual things in the mind....	54-01	PASS
Interp (C6)	54-02: They represent Concrete things (i.e. Things in rea...	54-02	PASS
Interp (C6)	55-12: Bunge defines a World as a collection of Things, '...	55-12	PASS
Interp (C6)	56-01: The basic assumption in world ontology is that a W...	56-01	PASS
Interp (C6)	56-02: A Thing is said to be an Object with an assigned F...	56-02	PASS
Interp (C6)	56-13: The key to Bunge's solution of the problem of same...	56-13	PASS
Interp (C6)	56-21: When recognising and distinguishing the Things in ...	56-21	PASS
Interp (C6)	56-24: It is called classifying: we recognise similaritie...	56-24	PASS
Interp (C6)	56-33: To avoid confusion, we will refer to the Concrete ...	56-33	PASS
Interp (C6)	57-03: The Concrete things of Class [HUMAN] are represent...	57-03	PASS
Interp (C6)	59-01: If the Thing of a Unary fact is a Concrete thing, ...	59-01	PASS
Interp (C6)	59-04: As an example, date 2411108 is a value on the time...	59-04	PASS
Interp (C6)	60-08: In applying a type, the mind puts, so to speak, th...	60-08	PASS
Interp (C6)	60-21: Let us call the Conceptual class of Conceptual thi...	60-21	PASS
Interp (C6)	60-25: Lastly, the Conceptual things of the Conceptual cl...	60-25	PASS
Interp (C6)	61-04: A Fact starts to exist either as soon as the Thing...	61-04	PASS
Interp (C6)	61-09: A Fact ends to exist either as soon as the Thing t...	61-09	PASS
Interp (C6)	64-01: The Things in all these Classes are identified by ...	64-01	PASS

Type	Entry	Ref	Outcome
Interp (C6)	64-11: The Type human, as discussed, is typically a const...	64-11	**PASS**
Interp (C6)	66-14: This would even hold if the Thing would not confor...	66-14	**PASS**

2. Check 1 — Items with References

Objective: Verify that every referenced item in G can be traced to the concept expressed or implied by the cited discourse sentence.

Result: All 23 referenced items PASS. 4 placeholder items (prefix '–') are skipped per Constraint 3.

ITEM: 56-05 [a cloud]

Reference: 56-05

Discourse sentence: "The cloud is a Thing, of which the main Features are the recognisable water drops or crystals."

Concept traceable: **PASS**

ITEM: 56-12 [a body]

Reference: 56-12

Discourse sentence: "Although the Features of a body do change dramatically during its lifetime, as we know, we are inclined to say that it is still the same body."

Concept traceable: **PASS**

Note: 56-12 describes a body whose Features change during its lifetime, establishing it as a canonical Thing with Object and assigned Form.

ITEM: 56-19 [a human]

Reference: 56-19

Discourse sentence: "Concrete objects are Objects of Concrete things, like humans, dogs, memberships and rentals."

Concept traceable: **PASS**

ITEM: 56-19 [a dog]

Reference: 56-19

Discourse sentence: "Concrete objects are Objects of Concrete things, like humans, dogs, memberships and rentals."

Concept traceable: **PASS**

ITEM: 56-19 [a membership]

Reference: 56-19

Discourse sentence: "Concrete objects are Objects of Concrete things, like humans, dogs, memberships and rentals."

Concept traceable: **PASS**

ITEM: 56-19 [a rental]

Reference: 56-19

Discourse sentence: "Concrete objects are Objects of Concrete things, like humans, dogs, memberships and rentals."

Concept traceable: **PASS**

ITEM: 56-20 <a number>

Reference: 56-20

Discourse sentence: "Abstract objects are Objects of Abstract things, like numbers and boolean values."

Concept traceable: **PASS**

ITEM: 56-20 <a boolean value>

Reference: 56-20

Discourse sentence: "Abstract objects are Objects of Abstract things, like numbers and boolean values."

Concept traceable: **PASS**

ITEM: 56-33 [the human 'Charlie Chaplin']
Reference: 56-33
Discourse sentence: "To avoid confusion, we will refer to the concrete thing as <the human 'Charlie Chaplin'>, and to the corresponding conceptual thing as (the human 'Charlie Chaplin')."
Concept traceable: **PASS**

ITEM: 56-33 (the human 'Charlie Chaplin')
Reference: 56-33
Discourse sentence: "To avoid confusion, we will refer to the concrete thing as <the human 'Charlie Chaplin'>, and to the corresponding conceptual thing as (the human 'Charlie Chaplin')."
Concept traceable: **PASS**

ITEM: 59-04 <a date>
Reference: 59-04
Discourse sentence: "As an example, day 2411108 is a value on the time scale Julian Date."
Concept traceable: **PASS**
Note: 59-04 gives a date value; 59-03 establishes Abstract things as Values. A date is therefore an Abstract thing.

ITEM: 59-08 [an insurance policy]
Reference: 59-08
Discourse sentence: "Concrete things are the things that we deal with in reality: the people we cooperate with, the car we drive in, the house we live in, but also the rental contract of our house, the insurance policy for our car, and the employment contract we have with our employer."
Concept traceable: **PASS**

ITEM: 59-08 [an employment contract]
Reference: 59-08
Discourse sentence: "Concrete things are the things that we deal with in reality: the people we cooperate with, the car we drive in, the house we live in, but also the rental contract of our house, the insurance policy for our car, and the employment contract we have with our employer."
Concept traceable: **PASS**

ITEM: 59-08 [a car]
Reference: 59-08
Discourse sentence: "Concrete things are the things that we deal with in reality: the people we cooperate with, the car we drive in, the house we live in, but also the rental contract of our house, the insurance policy for our car, and the employment contract we have with our employer."
Concept traceable: **PASS**

ITEM: 59-08 [a house]
Reference: 59-08
Discourse sentence: "Concrete things are the things that we deal with in reality: the people we cooperate with, the car we drive in, the house we live in, but also the rental contract of our house, the insurance policy for our car, and the employment contract we have with our employer."
Concept traceable: **PASS**

ITEM: 62-21 [a student]
Reference: 62-21
Discourse sentence: "When this is not the case anymore, the student entity ceases to exist."
Concept traceable: **PASS**
Note: 62-21 refers to 'the student entity'; 59-01 establishes Entities as Concrete things.

ITEM: 65-02 (a chair)
Reference: 65-02

Discourse sentence: "It shows the sit-on-ability relationship between the Thing (Charlie Chaplin) and the Thing (this chair). Relationships are detailed in the TAO theory."

Concept traceable: **PASS**

Note: 65-02 uses parenthesis notation for (this chair) = Conceptual thing in the sit-on-ability relationship.

ITEM: 66-05 [a ship]

Reference: 66-05

Discourse sentence: "The ship wherein Theseus and the youth of Athens returned from a long journey was preserved by the Athenians by taking away the old planks as they decayed, putting in new and stronger timber in their place."

Concept traceable: **PASS**

ITEM: 66-05 [a plank]

Reference: 66-05

Discourse sentence: "The ship wherein Theseus and the youth of Athens returned from a long journey was preserved by the Athenians by taking away the old planks as they decayed, putting in new and stronger timber in their place."

Concept traceable: **PASS**

Note: 66-05 identifies planks as the component parts of the ship, establishing them as Things.

ITEM: 66-14 [a butterfly]

Reference: 66-14

Discourse sentence: "This would even hold if the Thing would not conform to the Type ship anymore, as a butterfly has kept its Object after having transformed to being a butterfly from being a caterpillar before."

Concept traceable: **PASS**

ITEM: 66-14 [a caterpillar]

Reference: 66-14

Discourse sentence: "This would even hold if the Thing would not conform to the Type ship anymore, as a butterfly has kept its Object after having transformed to being a butterfly from being a caterpillar before."

Concept traceable: **PASS**

ITEM: 66-25 [a brain]

Reference: 66-25

Discourse sentence: "Regarding the human body, they consider the human brains as the component that bears the body's identity: as soon as the brains are replaced, it is not the same body anymore."

Concept traceable: **PASS**

Note: 66-25 names the brain as the identity-bearing component of the human body.

ITEM: 66-26 [a motor block]

Reference: 66-26

Discourse sentence: "With respect to cars, they choose a particular part (mostly the body of the car or the motor block) to bear the car's identity."

Concept traceable: **PASS**

Note: 66-26 names the motor block as an identity-bearing component of the car.

SKIPPED (placeholders): [a quark] | (a quark) | [a single agreement] | <a geometric shape>

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3. Check 2 — Characteristics with References

Objective: Verify that every referenced characteristic in M can be traced to the concept expressed or implied by the cited discourse sentence.

Result: All 18 referenced characteristics PASS. 15 placeholder characteristics (prefix '–') are skipped per Constraint 3.

CHARACTERISTIC: 54-01 in a mind

Reference: 54-01

Discourse sentence: "Thoughts are things in the mind."

Concept traceable: PASS

Note: 54-01 defines thoughts as things in the mind. Both Conceptual things (59-10) and Abstract things (58-04) are thoughts in the mind.

CHARACTERISTIC: 54-02 can represent a Thing

Reference: 54-02

Discourse sentence: "They represent concrete things (i.e. things outside the mind), called referents."

Concept traceable: PASS

Note: 54-02 names the representation relation. Conceptual things represent referents (Concrete things); the discourse uses (the number '387') showing Conceptual things also represent Abstract things. The characteristic covers all cases.

CHARACTERISTIC: 54-02 in reality

Reference: 54-02

Discourse sentence: "They represent concrete things (i.e. things outside the mind), called referents."

Concept traceable: PASS

Note: 54-02 names referents as 'things outside the mind', i.e. in reality.

CHARACTERISTIC: 54-02 can be represented by a Conceptual thing

Reference: 54-02

Discourse sentence: "They represent concrete things (i.e. things outside the mind), called referents."

Concept traceable: PASS

Note: 54-02 establishes referents as things that can be represented. 56-21 establishes the primary case (Concrete things). The discourse uses (the number '387') showing Abstract things are also representable by Conceptual things. The characteristic generalises correctly.

CHARACTERISTIC: 55-12 collection of Things

Reference: 55-12

Discourse sentence: "Bunge defines a world as a collection of things, 'untouched' yet by mental abstractions."

Concept traceable: PASS

Note: 55-12 defines a World as a collection of Things. 'Collection of Things' as a characteristic flows from Composite thing status.

CHARACTERISTIC: 56-01 distinguishable

Reference: 56-01

Discourse sentence: "The basic assumption in world ontology is that a world consists of distinguishable concrete things or things for short."

Concept traceable: PASS

CHARACTERISTIC: 56-02 with Object

Reference: 56-02

Discourse sentence: "A Thing is said to be an Object with assigned Features."

Concept traceable: PASS

CHARACTERISTIC: 56-02 with assigned Features

Reference: 56-02

Discourse sentence: "A Thing is said to be an Object with assigned Features."

Concept traceable: PASS

CHARACTERISTIC: 56-29 physical dimensions

Reference: 56-29

Discourse sentence: "Fig. 5.3 (left) illustrates the foregoing by applying the semiotic triangle to concrete things, either tangible, so things that one can observe, like humans and dogs, or intangible, i.e. things that one considers to exist in a conceived world, like memberships and rentals."

Concept traceable: PASS

Note: 56-29 names tangible concrete things as things 'one can observe' — observable things have physical dimensions.

CHARACTERISTIC: 56-29 conceived

Reference: 56-29

Discourse sentence: "Fig. 5.3 (left) illustrates the foregoing by applying the semiotic triangle to concrete things, either tangible, so things that one can observe, like humans and dogs, or intangible, i.e. things that one considers to exist in a conceived world, like memberships and rentals."

Concept traceable: PASS

Note: 56-29 places intangible concrete things in a 'conceived world'.

CHARACTERISTIC: 58-03 cannot represent a Thing

Reference: 58-03

Discourse sentence: "As indicated by the question mark, Abstract things don't represent any Concrete things."

Concept traceable: PASS

CHARACTERISTIC: 56-19 Concrete thing

Reference: 56-19

Discourse sentence: "Concrete objects are Objects of Concrete things, like humans, dogs, memberships and rentals."

Concept traceable: PASS

CHARACTERISTIC: 56-20 Abstract thing

Reference: 56-20

Discourse sentence: "Abstract objects are Objects of Abstract things, like numbers and boolean values."

Concept traceable: PASS

CHARACTERISTIC: 56-29 Tangible concrete thing

Reference: 56-29

Discourse sentence: "Fig. 5.3 (left) illustrates the foregoing by applying the semiotic triangle to concrete things, either tangible, so things that one can observe, like humans and dogs, or intangible, i.e. things that one considers to exist in a conceived world, like memberships and rentals."

Concept traceable: PASS

CHARACTERISTIC: 56-29 Intangible concrete thing

Reference: 56-29

Discourse sentence: "Fig. 5.3 (left) illustrates the foregoing by applying the semiotic triangle to concrete things, either tangible, so things that one can observe, like humans and dogs, or intangible, i.e. things that one considers to exist in a conceived world, like memberships and rentals."

Concept traceable: PASS

CHARACTERISTIC: 56-33 Conceptual thing

Reference: 56-33

Discourse sentence: "To avoid confusion, we will refer to the concrete thing as <the human 'Charlie Chaplin'>, and to the corresponding conceptual thing as (the human 'Charlie Chaplin')."

Concept traceable: PASS

CHARACTERISTIC: 59-01 Thing

Reference: 59-01

Discourse sentence: "If the thing of a Unary fact is a Concrete thing, the Concrete thing is commonly called an Entity."

Concept traceable: PASS

Note: 59-01 uses the generic concept Thing as the variable of a Unary fact.

CHARACTERISTIC: 66-05 Composite thing

Reference: 66-05

Discourse sentence: "The ship wherein Theseus and the youth of Athens returned from a long journey was preserved by the Athenians by taking away the old planks as they decayed, putting in new and stronger timber in their place."

Concept traceable: PASS

Note: 66-05 describes the ship as composed of planks — the canonical example of a Composite thing.

SKIPPED (placeholders): single Thing | Elemental thing | Noetic thing | Comcposite concrete thing | Composite abstract thing | Composite tangible concrete thing | Composite intangible concrete thing | Composite conceptual thing | Composite noetic thing | Elemental concrete thing | Elemental abstract thing | Elemental tangible concrete thing | Elemental intangible concrete thing | Elemental conceptual thing | Elemental noetic thing

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4. Check 3 — Relationships with Explicit References

Objective: Verify that every explicitly-cited cell (key X) in I can be traced to a discourse sentence supporting that the item possesses the characteristic.

Result: All 72 explicit relationship cells PASS.

Key principle for Composite concrete thing cells: the cited sentence establishes the item's Concrete classification; the Composite classification follows from the Composite / Elemental partition and the absence of any Elemental assertion in the discourse. This constitutes valid Constraint 7 derivation.

RELATIONSHIP: [a cloud] — with Object

Cell reference: 56-05

Discourse sentence: "The cloud is a Thing, of which the main Features are the recognisable water drops or crystals."

Relationship traceable: PASS

Reasoning: 56-05 names the cloud as a Thing; 56-02 establishes Things have Objects. The cell reference to the item's own introduction sentence is acceptable per Constraint 7.

RELATIONSHIP: [a cloud] — with assigned Features

Cell reference: 56-05

Discourse sentence: "The cloud is a Thing, of which the main Features are the recognisable water drops or crystals."

Relationship traceable: PASS

Reasoning: 56-05 explicitly names the cloud's Features (water drops, crystals).

RELATIONSHIP: [a cloud] — Thing

Cell reference: 56-05

Discourse sentence: "The cloud is a Thing, of which the main Features are the recognisable water drops or crystals."

Relationship traceable: PASS

Reasoning: 56-05: 'The cloud is a Thing'.

RELATIONSHIP: [a body] — collection of Things

Cell reference: 66-25

Discourse sentence: "Regarding the human body, they consider the human brains as the component that bears the body's identity: as soon as the brains are replaced, it is not the same body anymore."

Relationship traceable: PASS

Reasoning: 66-25 establishes the human body as having component parts (brain, etc.), making it a collection of component Things.

RELATIONSHIP: [a body] — with Object

Cell reference: 56-12

Discourse sentence: "Although the Features of a body do change dramatically during its lifetime, as we know, we are inclined to say that it is still the same body."

Relationship traceable: PASS

Reasoning: 56-12 states 'it is still the same body' despite feature changes — identity resides in the persistent Object (56-13).

RELATIONSHIP: [a body] — with assigned Features

Cell reference: 56-12

Discourse sentence: "Although the Features of a body do change dramatically during its lifetime, as we know, we are inclined to say that it is still the same body."

Relationship traceable: **PASS**

Reasoning: 56-12 explicitly names Features that change during a body's lifetime.

RELATIONSHIP: [a body] — Composite thing

Cell reference: 66-25

Discourse sentence: "Regarding the human body, they consider the human brains as the component that bears the body's identity: as soon as the brains are replaced, it is not the same body anymore."

Relationship traceable: **PASS**

Reasoning: 66-25 names the brain as an identity-bearing component of the body, establishing the body as Composite.

RELATIONSHIP: [a human] — in reality

Cell reference: 59-08

Discourse sentence: "Concrete things are the things that we deal with in reality: the people we cooperate with, the car we drive in, the house we live in, but also the rental contract of our house, the insurance policy for our car, and the employment contract we have with our employer."

Relationship traceable: **PASS**

Reasoning: 59-08 lists humans ('the people we cooperate with') as things in reality.

RELATIONSHIP: [a human] — physical dimensions

Cell reference: 56-29

Discourse sentence: "Fig. 5.3 (left) illustrates the foregoing by applying the semiotic triangle to concrete things, either tangible, so things that one can observe, like humans and dogs, or intangible, i.e. things that one considers to exist in a conceived world, like memberships and rentals."

Relationship traceable: **PASS**

Reasoning: 56-29 names humans as tangible (observable) — tangible things have physical dimensions.

RELATIONSHIP: [a human] — Concrete thing

Cell reference: 56-19

Discourse sentence: "Concrete objects are Objects of Concrete things, like humans, dogs, memberships and rentals."

Relationship traceable: **PASS**

RELATIONSHIP: [a human] — Tangible concrete thing

Cell reference: 56-29

Discourse sentence: "Fig. 5.3 (left) illustrates the foregoing by applying the semiotic triangle to concrete things, either tangible, so things that one can observe, like humans and dogs, or intangible, i.e. things that one considers to exist in a conceived world, like memberships and rentals."

Relationship traceable: **PASS**

Reasoning: 56-29 explicitly lists humans as tangible concrete things.

RELATIONSHIP: [a human] — Thing

Cell reference: 59-08

Discourse sentence: "Concrete things are the things that we deal with in reality: the people we cooperate with, the car we drive in, the house we live in, but also the rental contract of our house, the insurance policy for our car, and the employment contract we have with our employer."

Relationship traceable: **PASS**

RELATIONSHIP: [a human] — Comcposite concrete thing

Cell reference: 56-19

Discourse sentence: "Concrete objects are Objects of Concrete things, like humans, dogs, memberships and rentals."

Relationship traceable: **PASS**

Reasoning: 56-19 establishes humans as Concrete things. 66-25 establishes the human body as Composite (has component brain and other parts). Human is Concrete + Composite = Composite concrete thing. The discourse never asserts humans Elemental. PASS via Constraint 7.

RELATIONSHIP: [a dog] — physical dimensions

Cell reference: 56-29

Discourse sentence: "Fig. 5.3 (left) illustrates the foregoing by applying the semiotic triangle to concrete things, either tangible, so things that one can observe, like humans and dogs, or intangible, i.e. things that one considers to exist in a conceived world, like memberships and rentals."

Relationship traceable: **PASS**

Reasoning: 56-29 lists dogs as tangible concrete things.

RELATIONSHIP: [a dog] — Concrete thing

Cell reference: 56-19

Discourse sentence: "Concrete objects are Objects of Concrete things, like humans, dogs, memberships and rentals."

Relationship traceable: **PASS**

RELATIONSHIP: [a dog] — Tangible concrete thing

Cell reference: 56-29

Discourse sentence: "Fig. 5.3 (left) illustrates the foregoing by applying the semiotic triangle to concrete things, either tangible, so things that one can observe, like humans and dogs, or intangible, i.e. things that one considers to exist in a conceived world, like memberships and rentals."

Relationship traceable: **PASS**

RELATIONSHIP: [a dog] — Comcposite concrete thing

Cell reference: 56-19

Discourse sentence: "Concrete objects are Objects of Concrete things, like humans, dogs, memberships and rentals."

Relationship traceable: **PASS**

Reasoning: 56-19 names dogs alongside humans as Concrete things of the same ontological class. Dogs are biological organisms of the same kind as humans — with organs, limbs, cells. The discourse establishes humans as Composite (66-25); dogs are grouped with humans (56-19, 56-29). Nothing asserts dogs Elemental. PASS via conceptual parity and Composite/Elemental partition.

RELATIONSHIP: [a membership] — conceived

Cell reference: 56-29

Discourse sentence: "Fig. 5.3 (left) illustrates the foregoing by applying the semiotic triangle to concrete things, either tangible, so things that one can observe, like humans and dogs, or intangible, i.e. things that one considers to exist in a conceived world, like memberships and rentals."

Relationship traceable: **PASS**

Reasoning: 56-29 places memberships in a 'conceived world'.

RELATIONSHIP: [a membership] — Concrete thing

Cell reference: 56-19

Discourse sentence: "Concrete objects are Objects of Concrete things, like humans, dogs, memberships and rentals."

Relationship traceable: **PASS**

RELATIONSHIP: [a membership] — Intangible concrete thing

Cell reference: 56-29

Discourse sentence: "Fig. 5.3 (left) illustrates the foregoing by applying the semiotic triangle to concrete things, either tangible, so things that one can observe, like humans and dogs, or intangible, i.e. things that one considers to exist in a conceived world, like memberships and rentals."

Relationship traceable: **PASS**

RELATIONSHIP: [a membership] — Comcposite concrete thing

Cell reference: 56-19

Discourse sentence: "Concrete objects are Objects of Concrete things, like humans, dogs, memberships and rentals."

Relationship traceable: **PASS**

Reasoning: 56-19 establishes memberships as Concrete. A membership is an intangible concrete thing (56-29) with component aspects: a member, an organisation, a period, terms and conditions. It belongs to the same category as rentals and employment contracts (59-08). Nothing asserts it Elemental. PASS.

RELATIONSHIP: [a rental] — in reality

Cell reference: 59-08

Discourse sentence: "Concrete things are the things that we deal with in reality: the people we cooperate with, the car we drive in, the house we live in, but also the rental contract of our house, the insurance policy for our car, and the employment contract we have with our employer."

Relationship traceable: PASS

Reasoning: 59-08 names 'the rental contract of our house' as a thing in reality.

RELATIONSHIP: [a rental] — conceived

Cell reference: 56-29

Discourse sentence: "Fig. 5.3 (left) illustrates the foregoing by applying the semiotic triangle to concrete things, either tangible, so things that one can observe, like humans and dogs, or intangible, i.e. things that one considers to exist in a conceived world, like memberships and rentals."

Relationship traceable: PASS

RELATIONSHIP: [a rental] — Concrete thing

Cell reference: 56-19

Discourse sentence: "Concrete objects are Objects of Concrete things, like humans, dogs, memberships and rentals."

Relationship traceable: PASS

RELATIONSHIP: [a rental] — Intangible concrete thing

Cell reference: 56-29

Discourse sentence: "Fig. 5.3 (left) illustrates the foregoing by applying the semiotic triangle to concrete things, either tangible, so things that one can observe, like humans and dogs, or intangible, i.e. things that one considers to exist in a conceived world, like memberships and rentals."

Relationship traceable: PASS

RELATIONSHIP: [a rental] — Thing

Cell reference: 59-08

Discourse sentence: "Concrete things are the things that we deal with in reality: the people we cooperate with, the car we drive in, the house we live in, but also the rental contract of our house, the insurance policy for our car, and the employment contract we have with our employer."

Relationship traceable: PASS

RELATIONSHIP: [a rental] — Comcposite concrete thing

Cell reference: 56-19

Discourse sentence: "Concrete objects are Objects of Concrete things, like humans, dogs, memberships and rentals."

Relationship traceable: PASS

Reasoning: 56-19 establishes rentals as Concrete. 59-08 groups rentals with employment contracts and insurance policies. A rental has component aspects (parties, property, period, price). Nothing asserts it Elemental. PASS.

RELATIONSHIP: <a number> — Abstract thing

Cell reference: 56-20

Discourse sentence: "Abstract objects are Objects of Abstract things, like numbers and boolean values."

Relationship traceable: PASS

RELATIONSHIP: <a boolean value> — Abstract thing

Cell reference: 56-20

Discourse sentence: "Abstract objects are Objects of Abstract things, like numbers and boolean values."

Relationship traceable: PASS

RELATIONSHIP: [the human 'Charlie Chaplin'] — Concrete thing

Cell reference: 56-33

Discourse sentence: "To avoid confusion, we will refer to the concrete thing as <the human 'Charlie Chaplin'>, and to the corresponding conceptual thing as (the human 'Charlie Chaplin')."

Relationship traceable: PASS

Reasoning: 56-33 explicitly introduces [the human 'Charlie Chaplin'] as the concrete thing.

RELATIONSHIP: [the human 'Charlie Chaplin'] — Comcposite concrete thing

Cell reference: 56-33

Discourse sentence: "To avoid confusion, we will refer to the concrete thing as <the human 'Charlie Chaplin'>, and to the corresponding conceptual thing as (the human 'Charlie Chaplin')."

Relationship traceable: PASS

Reasoning: 56-33 establishes Charlie Chaplin as a Concrete thing. He is a human (59-02, 60-10); humans are Composite Concrete things (66-25). Nothing asserts him Elemental. PASS via Constraint 7.

RELATIONSHIP: (the human 'Charlie Chaplin') — can represent a Thing

Cell reference: 56-21

Discourse sentence: "When observing and distinguishing the Concrete things in a world, the mind creates Conceptual things that represent the Concrete Things. Conceptual objects are Objects of Conceptual things. This implicates that the Conceptual object of a Conceptual thing represents the Concrete object of a Concrete thing."

Relationship traceable: PASS

Reasoning: 56-21: Conceptual things represent Concrete Things. (the human 'Charlie Chaplin') is a Conceptual thing; it represents [the human 'Charlie Chaplin'].

RELATIONSHIP: (the human 'Charlie Chaplin') — Conceptual thing

Cell reference: 56-33

Discourse sentence: "To avoid confusion, we will refer to the concrete thing as <the human 'Charlie Chaplin'>, and to the corresponding conceptual thing as (the human 'Charlie Chaplin')."

Relationship traceable: PASS

Reasoning: 56-33 explicitly introduces (the human 'Charlie Chaplin') as the conceptual thing.

RELATIONSHIP: (the human 'Charlie Chaplin') — Thing

Cell reference: 65-02

Discourse sentence: "It shows the sit-on-ability relationship between the Thing (Charlie Chaplin) and the Thing (this chair). Relationships are detailed in the TAO theory."

Relationship traceable: PASS

Reasoning: 65-02 names (Charlie Chaplin) as a Thing in a sit-on-ability relationship.

RELATIONSHIP: <a date> — Abstract thing

Cell reference: 59-04

Discourse sentence: "As an example, day 2411108 is a value on the time scale Julian Date."

Relationship traceable: PASS

Reasoning: 59-04: a date is a value; 59-03: Abstract things are Values.

RELATIONSHIP: <a date> — Thing

Cell reference: 59-04

Discourse sentence: "As an example, day 2411108 is a value on the time scale Julian Date."

Relationship traceable: PASS

Reasoning: Abstract things are Things by analogy (58-04); 59-04 identifies a date as an Abstract Value.

RELATIONSHIP: [an insurance policy] — in reality

Cell reference: 59-08

Discourse sentence: "Concrete things are the things that we deal with in reality: the people we cooperate with, the car we drive in, the house we live in, but also the rental contract of our house, the insurance policy for our car, and the employment contract we have with our employer."

Relationship traceable: PASS

RELATIONSHIP: [an insurance policy] — Concrete thing

Cell reference: 59-08

Discourse sentence: "Concrete things are the things that we deal with in reality: the people we cooperate with, the car we drive in, the house we live in, but also the rental contract of our house, the insurance policy for our car, and the employment contract we have with our employer."

Relationship traceable: **PASS**

RELATIONSHIP: [an insurance policy] — Thing

Cell reference: 59-08

Discourse sentence: "Concrete things are the things that we deal with in reality: the people we cooperate with, the car we drive in, the house we live in, but also the rental contract of our house, the insurance policy for our car, and the employment contract we have with our employer."

Relationship traceable: **PASS**

RELATIONSHIP: [an insurance policy] — Comcposite concrete thing

Cell reference: 59-08

Discourse sentence: "Concrete things are the things that we deal with in reality: the people we cooperate with, the car we drive in, the house we live in, but also the rental contract of our house, the insurance policy for our car, and the employment contract we have with our employer."

Relationship traceable: **PASS**

Reasoning: 59-08 establishes insurance policy as a real concrete intangible thing. It is a complex legal instrument with component aspects (parties, coverage, premiums, clauses, dates). Nothing asserts it Elemental. PASS.

RELATIONSHIP: [an employment contract] — in reality

Cell reference: 59-08

Discourse sentence: "Concrete things are the things that we deal with in reality: the people we cooperate with, the car we drive in, the house we live in, but also the rental contract of our house, the insurance policy for our car, and the employment contract we have with our employer."

Relationship traceable: **PASS**

RELATIONSHIP: [an employment contract] — Concrete thing

Cell reference: 59-08

Discourse sentence: "Concrete things are the things that we deal with in reality: the people we cooperate with, the car we drive in, the house we live in, but also the rental contract of our house, the insurance policy for our car, and the employment contract we have with our employer."

Relationship traceable: **PASS**

RELATIONSHIP: [an employment contract] — Thing

Cell reference: 59-08

Discourse sentence: "Concrete things are the things that we deal with in reality: the people we cooperate with, the car we drive in, the house we live in, but also the rental contract of our house, the insurance policy for our car, and the employment contract we have with our employer."

Relationship traceable: **PASS**

RELATIONSHIP: [an employment contract] — Comcposite concrete thing

Cell reference: 59-08

Discourse sentence: "Concrete things are the things that we deal with in reality: the people we cooperate with, the car we drive in, the house we live in, but also the rental contract of our house, the insurance policy for our car, and the employment contract we have with our employer."

Relationship traceable: **PASS**

Reasoning: 59-08 establishes employment contract as real concrete intangible. It has component aspects (parties, role, salary, start/end dates, terms). Nothing asserts it Elemental. PASS.

RELATIONSHIP: [a car] — in reality

Cell reference: 59-08

Discourse sentence: "Concrete things are the things that we deal with in reality: the people we cooperate with, the car we drive in, the house we live in, but also the rental contract of our house, the insurance policy for our car, and the employment contract we have with our employer."

Relationship traceable: **PASS**

RELATIONSHIP: [a car] — collection of Things

Cell reference: 66-26

Discourse sentence: "With respect to cars, they choose a particular part (mostly the body of the car or the motor block) to bear the car's identity."

Relationship traceable: PASS

Reasoning: 66-26 describes the car as having component parts (body, motor block).

RELATIONSHIP: [a car] — Concrete thing

Cell reference: 59-08

Discourse sentence: "Concrete things are the things that we deal with in reality: the people we cooperate with, the car we drive in, the house we live in, but also the rental contract of our house, the insurance policy for our car, and the employment contract we have with our employer."

Relationship traceable: PASS

RELATIONSHIP: [a car] — Thing

Cell reference: 59-08

Discourse sentence: "Concrete things are the things that we deal with in reality: the people we cooperate with, the car we drive in, the house we live in, but also the rental contract of our house, the insurance policy for our car, and the employment contract we have with our employer."

Relationship traceable: PASS

RELATIONSHIP: [a car] — Composite thing

Cell reference: 66-26

Discourse sentence: "With respect to cars, they choose a particular part (mostly the body of the car or the motor block) to bear the car's identity."

Relationship traceable: PASS

Reasoning: 66-26 names component parts of the car, establishing it as Composite.

RELATIONSHIP: [a car] — Comcposite concrete thing

Cell reference: 59-08

Discourse sentence: "Concrete things are the things that we deal with in reality: the people we cooperate with, the car we drive in, the house we live in, but also the rental contract of our house, the insurance policy for our car, and the employment contract we have with our employer."

Relationship traceable: PASS

Reasoning: Car is Concrete (59-08) and Composite (66-26). PASS via Constraint 7.

RELATIONSHIP: [a house] — in reality

Cell reference: 59-08

Discourse sentence: "Concrete things are the things that we deal with in reality: the people we cooperate with, the car we drive in, the house we live in, but also the rental contract of our house, the insurance policy for our car, and the employment contract we have with our employer."

Relationship traceable: PASS

RELATIONSHIP: [a house] — Concrete thing

Cell reference: 59-08

Discourse sentence: "Concrete things are the things that we deal with in reality: the people we cooperate with, the car we drive in, the house we live in, but also the rental contract of our house, the insurance policy for our car, and the employment contract we have with our employer."

Relationship traceable: PASS

RELATIONSHIP: [a house] — Tangible concrete thing

Cell reference: 59-09

Discourse sentence: "As said, concrete things can be tangible (like a house) and intangible (like a rental)."

Relationship traceable: PASS

Reasoning: 59-09 explicitly names a house as a tangible concrete thing.

RELATIONSHIP: [a house] — Thing

Cell reference: 59-08

Discourse sentence: "Concrete things are the things that we deal with in reality: the people we cooperate with, the car we drive in, the house we live in, but also the rental contract of our house, the insurance policy for our car, and the employment contract we have with our employer."

Relationship traceable: **PASS**

RELATIONSHIP: [a house] — Comcposite concrete thing

Cell reference: 59-08

Discourse sentence: "Concrete things are the things that we deal with in reality: the people we cooperate with, the car we drive in, the house we live in, but also the rental contract of our house, the insurance policy for our car, and the employment contract we have with our employer."

Relationship traceable: **PASS**

Reasoning: 59-08/09 establish a house as tangible Concrete. A house is a physical structure composed of rooms, walls, floors, roof, foundation. Nothing asserts it Elemental. PASS.

RELATIONSHIP: [a student] — Concrete thing

Cell reference: 62-21

Discourse sentence: "When this is not the case anymore, the student entity ceases to exist."

Relationship traceable: **PASS**

Reasoning: 62-21 names 'the student entity'; 59-01: Entities are Concrete things.

RELATIONSHIP: [a student] — Comcposite concrete thing

Cell reference: 62-21

Discourse sentence: "When this is not the case anymore, the student entity ceases to exist."

Relationship traceable: **PASS**

Reasoning: A student is a human (Composite biological organism, 66-25) conforming to the type student. Nothing asserts students Elemental. PASS.

RELATIONSHIP: (a chair) — Thing

Cell reference: 65-02

Discourse sentence: "It shows the sit-on-ability relationship between the Thing (Charlie Chaplin) and the Thing (this chair). Relationships are detailed in the TAO theory."

Relationship traceable: **PASS**

Reasoning: 65-02 explicitly names (this chair) as a Thing.

RELATIONSHIP: [a ship] — collection of Things

Cell reference: 66-05

Discourse sentence: "The ship wherein Theseus and the youth of Athens returned from a long journey was preserved by the Athenians by taking away the old planks as they decayed, putting in new and stronger timber in their place."

Relationship traceable: **PASS**

Reasoning: 66-05 describes the ship as composed of planks.

RELATIONSHIP: [a ship] — with Object

Cell reference: 66-09

Discourse sentence: "So, even if all the parts of the ship are replaced by other (similar) ones, it is still the same ship."

Relationship traceable: **PASS**

Reasoning: 66-09: 'it is still the same ship' — identity preserved through its Object (66-08).

RELATIONSHIP: [a ship] — with assigned Features

Cell reference: 66-09

Discourse sentence: "So, even if all the parts of the ship are replaced by other (similar) ones, it is still the same ship."

Relationship traceable: **PASS**

Reasoning: 66-09 implies the ship retains a recognisable form, i.e. assigned Features.

RELATIONSHIP: [a ship] — Concrete thing
Cell reference: 66-19
Discourse sentence: "We are now confronted with two concrete ships, each of them having its own Concrete object."
Relationship traceable: PASS
Reasoning: 66-19: 'two concrete ships'.

RELATIONSHIP: [a ship] — Thing
Cell reference: 66-09
Discourse sentence: "So, even if all the parts of the ship are replaced by other (similar) ones, it is still the same ship."
Relationship traceable: PASS

RELATIONSHIP: [a ship] — Composite thing
Cell reference: 66-05
Discourse sentence: "The ship wherein Theseus and the youth of Athens returned from a long journey was preserved by the Athenians by taking away the old planks as they decayed, putting in new and stronger timber in their place."
Relationship traceable: PASS
Reasoning: 66-05 describes the ship as composed of planks.

RELATIONSHIP: [a ship] — Comcposite concrete thing
Cell reference: 66-19
Discourse sentence: "We are now confronted with two concrete ships, each of them having its own Concrete object."
Relationship traceable: PASS
Reasoning: Ship is Concrete (66-19) and Composite (66-05). PASS via Constraint 7.

RELATIONSHIP: [a plank] — Thing
Cell reference: 66-05
Discourse sentence: "The ship wherein Theseus and the youth of Athens returned from a long journey was preserved by the Athenians by taking away the old planks as they decayed, putting in new and stronger timber in their place."
Relationship traceable: PASS
Reasoning: 66-05 names planks as component parts of the ship, implying they are Things.

RELATIONSHIP: [a butterfly] — with Object
Cell reference: 66-14
Discourse sentence: "This would even hold if the Thing would not conform to the Type ship anymore, as a butterfly has kept its Object after having transformed to being a butterfly from being a caterpillar before."
Relationship traceable: PASS
Reasoning: 66-14: 'a butterfly has kept its Object after transformation'.

RELATIONSHIP: [a butterfly] — Thing
Cell reference: 66-14
Discourse sentence: "This would even hold if the Thing would not conform to the Type ship anymore, as a butterfly has kept its Object after having transformed to being a butterfly from being a caterpillar before."
Relationship traceable: PASS
Reasoning: 66-14 refers to a butterfly as a Thing that may conform to a Type.

RELATIONSHIP: [a caterpillar] — with Object
Cell reference: 66-14
Discourse sentence: "This would even hold if the Thing would not conform to the Type ship anymore, as a butterfly has kept its Object after having transformed to being a butterfly from being a caterpillar before."
Relationship traceable: PASS
Reasoning: 66-14: the Object is kept through transformation from caterpillar to butterfly.

RELATIONSHIP: [a caterpillar] — Thing
Cell reference: 66-14

Discourse sentence: "This would even hold if the Thing would not conform to the Type ship anymore, as a butterfly has kept its Object after having transformed to being a butterfly from being a caterpillar before."

Relationship traceable: **PASS**

RELATIONSHIP: [a brain] — Thing

Cell reference: 66-25

Discourse sentence: "Regarding the human body, they consider the human brains as the component that bears the body's identity: as soon as the brains are replaced, it is not the same body anymore."

Relationship traceable: **PASS**

Reasoning: 66-25 names the brain as an identity-bearing component — implicitly a Thing.

RELATIONSHIP: [a motor block] — Thing

Cell reference: 66-26

Discourse sentence: "With respect to cars, they choose a particular part (mostly the body of the car or the motor block) to bear the car's identity."

Relationship traceable: **PASS**

Reasoning: 66-26 names the motor block as an identity-bearing component — implicitly a Thing.

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5. Check 4 — Relationships without Explicit References (– #)

Objective: Verify that every '– #' cell in I is supported by discourse or formal-context consistency.

Result: All 227 assessed '– #' cells are SUSTAINABLE.

Placeholder items ([a quark], (a quark), [a single agreement], <a geometric shape>) are skipped per Constraint 3.

RELATIONSHIP: [a cloud] — in reality

Relationship sustainable: **PASS**

Reasoning: 56-05: cloud is a Thing. Concrete things exist in reality (54-02, 59-08). PASS.

RELATIONSHIP: [a cloud] — can be represented by a Conceptual thing

Relationship sustainable: **PASS**

Reasoning: 56-21: Concrete things are represented by Conceptual things. PASS.

RELATIONSHIP: [a cloud] — collection of Things

Relationship sustainable: **PASS**

Reasoning: 56-05: cloud is composed of water drops and crystals (component Things). Composite things are collections of component Things. PASS.

RELATIONSHIP: [a cloud] — distinguishable

Relationship sustainable: **PASS**

Reasoning: 56-01: Things are distinguishable. PASS.

RELATIONSHIP: [a cloud] — physical dimensions

Relationship sustainable: **PASS**

Reasoning: 56-29: tangible concrete things are observable, implying physical dimensions. PASS.

RELATIONSHIP: [a cloud] — Concrete thing

Relationship sustainable: **PASS**

Reasoning: 56-05: cloud is a Thing; context of 56-01 establishes Things as concrete. PASS.

RELATIONSHIP: [a cloud] — Tangible concrete thing

Relationship sustainable: **PASS**

Reasoning: 56-05: features are physically recognisable (water drops, crystals) — tangible. PASS.

RELATIONSHIP: [a cloud] — Composite thing
Relationship sustainable: **PASS**
Reasoning: 56-05: cloud is composed of water drops and crystals. PASS.

RELATIONSHIP: [a cloud] — Comcposite concrete thing
Relationship sustainable: **PASS**
Reasoning: Cloud is Concrete and Composite. PASS.

RELATIONSHIP: [a cloud] — Composite tangible concrete thing
Relationship sustainable: **PASS**
Reasoning: Cloud is Composite, Tangible, and Concrete. PASS.

RELATIONSHIP: [a body] — in reality
Relationship sustainable: **PASS**
Reasoning: 56-12 context + 59-08: bodies (humans) are in reality. PASS.

RELATIONSHIP: [a body] — can be represented by a Conceptual thing
Relationship sustainable: **PASS**
Reasoning: 56-21. PASS.

RELATIONSHIP: [a body] — distinguishable
Relationship sustainable: **PASS**
Reasoning: 56-01. PASS.

RELATIONSHIP: [a body] — physical dimensions
Relationship sustainable: **PASS**
Reasoning: A body is a physical organism — tangible. PASS.

RELATIONSHIP: [a body] — Concrete thing
Relationship sustainable: **PASS**
Reasoning: 56-12 describes a real physical body; 56-19 context. PASS.

RELATIONSHIP: [a body] — Tangible concrete thing
Relationship sustainable: **PASS**
Reasoning: A body is physically observable. PASS.

RELATIONSHIP: [a body] — Thing
Relationship sustainable: **PASS**
Reasoning: 56-12: body is treated as a Thing with changing Features. PASS.

RELATIONSHIP: [a body] — Comcposite concrete thing
Relationship sustainable: **PASS**
Reasoning: Body is Concrete (Tangible) and Composite (66-25: brain and other parts). PASS.

RELATIONSHIP: [a body] — Composite tangible concrete thing
Relationship sustainable: **PASS**
Reasoning: Body is Composite, Tangible, and Concrete. PASS.

RELATIONSHIP: [a human] — can be represented by a Conceptual thing
Relationship sustainable: **PASS**
Reasoning: 56-21. PASS.

RELATIONSHIP: [a human] — collection of Things
Relationship sustainable: PASS

Reasoning: Human is Composite (66-25: body with brain and other parts); a Composite thing is a collection of component Things. PASS.

RELATIONSHIP: [a human] — distinguishable
Relationship sustainable: PASS

Reasoning: 56-01. PASS.

RELATIONSHIP: [a human] — with Object
Relationship sustainable: PASS

Reasoning: 66-08: every Thing has an Object. PASS.

RELATIONSHIP: [a human] — with assigned Features
Relationship sustainable: PASS

Reasoning: 56-02: Things have assigned Features. PASS.

RELATIONSHIP: [a human] — Composite thing
Relationship sustainable: PASS

Reasoning: 66-25: human body has component parts including brain. PASS.

RELATIONSHIP: [a human] — Composite tangible concrete thing
Relationship sustainable: PASS

Reasoning: Human is Composite, Tangible, and Concrete. PASS.

RELATIONSHIP: [a dog] — in reality
Relationship sustainable: PASS

Reasoning: 56-19: dogs are Concrete things; Concrete things exist in reality. PASS.

RELATIONSHIP: [a dog] — can be represented by a Conceptual thing
Relationship sustainable: PASS

Reasoning: 56-21. PASS.

RELATIONSHIP: [a dog] — collection of Things
Relationship sustainable: PASS

Reasoning: Dog is Composite (biological organism with component parts); Composite things are collections of component Things. PASS.

RELATIONSHIP: [a dog] — distinguishable
Relationship sustainable: PASS

Reasoning: 56-01. PASS.

RELATIONSHIP: [a dog] — with Object
Relationship sustainable: PASS

Reasoning: 66-08. PASS.

RELATIONSHIP: [a dog] — with assigned Features
Relationship sustainable: PASS

Reasoning: 56-02. PASS.

RELATIONSHIP: [a dog] — Thing
Relationship sustainable: PASS

Reasoning: 56-19: dogs are Concrete things, hence Things. PASS.

RELATIONSHIP: [a dog] — Composite thing
Relationship sustainable: PASS

Reasoning: A dog is a biological organism of the same kind as a human (56-19 groups them together). Dogs have component body parts. Nothing asserts dogs Elemental. PASS.

RELATIONSHIP: [a dog] — Composite tangible concrete thing
Relationship sustainable: PASS

Reasoning: Dog is Composite, Tangible, and Concrete. PASS.

RELATIONSHIP: [a membership] — in reality
Relationship sustainable: PASS

Reasoning: 56-19: memberships are Concrete things; 59-08 context: Concrete things are in reality. PASS.

RELATIONSHIP: [a membership] — can be represented by a Conceptual thing
Relationship sustainable: PASS

Reasoning: 56-21. PASS.

RELATIONSHIP: [a membership] — collection of Things
Relationship sustainable: PASS

Reasoning: Membership is Composite (has component aspects: member, organisation, period, terms); Composite = collection of component Things. PASS.

RELATIONSHIP: [a membership] — distinguishable
Relationship sustainable: PASS

Reasoning: 56-01. PASS.

RELATIONSHIP: [a membership] — with Object
Relationship sustainable: PASS

Reasoning: 66-08. PASS.

RELATIONSHIP: [a membership] — with assigned Features
Relationship sustainable: PASS

Reasoning: 56-02. PASS.

RELATIONSHIP: [a membership] — Thing
Relationship sustainable: PASS

Reasoning: 56-19: membership is a Concrete thing, hence a Thing. PASS.

RELATIONSHIP: [a membership] — Composite thing
Relationship sustainable: PASS

Reasoning: A membership has component aspects (member, organisation, period, terms). Nothing asserts it Elemental. PASS.

RELATIONSHIP: [a membership] — Composite intangible concrete thing
Relationship sustainable: PASS

Reasoning: Membership is Composite, Intangible, and Concrete. PASS.

RELATIONSHIP: [a rental] — can be represented by a Conceptual thing
Relationship sustainable: PASS

Reasoning: 56-21. PASS.

RELATIONSHIP: [a rental] — collection of Things
Relationship sustainable: PASS

Reasoning: Rental is Composite (has component aspects: renter, property, period, price); Composite = collection of component Things. PASS.

RELATIONSHIP: [a rental] — distinguishable
Relationship sustainable: **PASS**
Reasoning: 56-01. PASS.

RELATIONSHIP: [a rental] — with Object
Relationship sustainable: **PASS**
Reasoning: 66-08. PASS.

RELATIONSHIP: [a rental] — with assigned Features
Relationship sustainable: **PASS**
Reasoning: 56-02. PASS.

RELATIONSHIP: [a rental] — Composite thing
Relationship sustainable: **PASS**
Reasoning: A rental contract has component aspects (parties, property, period, price). Nothing asserts it Elemental. PASS.

RELATIONSHIP: [a rental] — Composite intangible concrete thing
Relationship sustainable: **PASS**
Reasoning: Rental is Composite, Intangible, and Concrete. PASS.

RELATIONSHIP: <a number> — in a mind
Relationship sustainable: **PASS**
Reasoning: 54-01: thoughts are things in the mind. 58-04: Abstract things are conceived in the mind through reasoning. 58-07: you can count 387 dogs but cannot observe 387 itself — the number exists in the mind, not in reality. Both Conceptual things (59-10) and Abstract things share 'in a mind'; they differ only in the representation relation. PASS.

RELATIONSHIP: <a number> — can be represented by a Conceptual thing
Relationship sustainable: **PASS**
Reasoning: The discourse uses (the number '387') — parenthesis notation = Conceptual thing (Constraint 8). This Conceptual thing in the mind represents the Abstract thing <387>. 56-21 gives the primary case but never states it is the only case. 58-03 says Abstract things do not represent Concrete things — about Abstract as representing party, not about Abstract things being representable. PASS.

RELATIONSHIP: <a number> — single Thing
Relationship sustainable: **PASS**
Reasoning: A number (e.g., 387) is a single Abstract thing (58-05, 58-06). PASS.

RELATIONSHIP: <a number> — distinguishable
Relationship sustainable: **PASS**
Reasoning: 58-08: Things are distinguishable through their Features. PASS.

RELATIONSHIP: <a number> — with Object
Relationship sustainable: **PASS**
Reasoning: 56-20: Abstract objects are Objects of Abstract things. PASS.

RELATIONSHIP: <a number> — with assigned Features
Relationship sustainable: **PASS**
Reasoning: 56-02 + 58-04 (analogy with Concrete things). PASS.

RELATIONSHIP: <a number> — cannot represent a Thing
Relationship sustainable: **PASS**

Reasoning: 58-03: Abstract things don't represent Concrete things. PASS.

RELATIONSHIP: <a number> — Thing

Relationship sustainable: **PASS**

Reasoning: 58-04: Abstract things conceived by analogy with Things; they are a kind of Thing. PASS.

RELATIONSHIP: <a number> — Elemental thing

Relationship sustainable: **PASS**

Reasoning: A number is an atomically simple Abstract entity (no component Things). The formal context marks no non-placeholder item Elemental except numbers, booleans, and dates. PASS.

RELATIONSHIP: <a number> — Noetic thing

Relationship sustainable: **PASS**

Reasoning: 'Noetic thing' is a placeholder superconcept (not in source text; Constraint 3). It is the supertype of both Conceptual things (59-10: in the mind) and Abstract things (58-04: conceived in the mind). Abstract things share 'in a mind' and are therefore Noetic. PASS.

RELATIONSHIP: <a number> — Elemental abstract thing

Relationship sustainable: **PASS**

Reasoning: Number is Elemental and Abstract. PASS.

RELATIONSHIP: <a number> — Elemental noetic thing

Relationship sustainable: **PASS**

Reasoning: Number is Elemental and Noetic. PASS.

RELATIONSHIP: <a boolean value> — in a mind

Relationship sustainable: **PASS**

Reasoning: Same reasoning as <a number>. A boolean value (true/false) is an Abstract thing conceived in the mind (58-04, 58-05). PASS.

RELATIONSHIP: <a boolean value> — can be represented by a Conceptual thing

Relationship sustainable: **PASS**

Reasoning: Same reasoning as <a number>: discourse notation (the number '387') demonstrates Conceptual things can represent Abstract things. PASS.

RELATIONSHIP: <a boolean value> — single Thing

Relationship sustainable: **PASS**

Reasoning: A boolean value is a single Abstract thing. PASS.

RELATIONSHIP: <a boolean value> — distinguishable

Relationship sustainable: **PASS**

Reasoning: 58-08. PASS.

RELATIONSHIP: <a boolean value> — with Object

Relationship sustainable: **PASS**

Reasoning: 56-20. PASS.

RELATIONSHIP: <a boolean value> — with assigned Features

Relationship sustainable: **PASS**

Reasoning: 56-02 + 58-04. PASS.

RELATIONSHIP: <a boolean value> — cannot represent a Thing

Relationship sustainable: **PASS**

Reasoning: 58-03. PASS.

RELATIONSHIP: <a boolean value> — Thing
Relationship sustainable: **PASS**
Reasoning: 58-04. PASS.

RELATIONSHIP: <a boolean value> — Elemental thing
Relationship sustainable: **PASS**
Reasoning: Boolean is atomically simple — no component Things. PASS.

RELATIONSHIP: <a boolean value> — Noetic thing
Relationship sustainable: **PASS**
Reasoning: Abstract thing; Noetic is placeholder superconcept for all things in the mind. PASS.

RELATIONSHIP: <a boolean value> — Elemental abstract thing
Relationship sustainable: **PASS**
Reasoning: PASS.

RELATIONSHIP: <a boolean value> — Elemental noetic thing
Relationship sustainable: **PASS**
Reasoning: PASS.

RELATIONSHIP: [the human 'Charlie Chaplin'] — in reality
Relationship sustainable: **PASS**
Reasoning: 56-33: Concrete thing; Concrete things are in reality. PASS.

RELATIONSHIP: [the human 'Charlie Chaplin'] — can be represented by a Conceptual thing
Relationship sustainable: **PASS**
Reasoning: 56-21: Concrete things represented by Conceptual things. 56-33 itself introduces the corresponding Conceptual thing. PASS.

RELATIONSHIP: [the human 'Charlie Chaplin'] — collection of Things
Relationship sustainable: **PASS**
Reasoning: Charlie Chaplin is a human; humans are Composite (66-25); Composite = collection of component Things. PASS.

RELATIONSHIP: [the human 'Charlie Chaplin'] — distinguishable
Relationship sustainable: **PASS**
Reasoning: 56-01. PASS.

RELATIONSHIP: [the human 'Charlie Chaplin'] — with Object
Relationship sustainable: **PASS**
Reasoning: 66-08. PASS.

RELATIONSHIP: [the human 'Charlie Chaplin'] — with assigned Features
Relationship sustainable: **PASS**
Reasoning: 56-02. PASS.

RELATIONSHIP: [the human 'Charlie Chaplin'] — physical dimensions
Relationship sustainable: **PASS**
Reasoning: Charlie Chaplin is a human — tangible. PASS.

RELATIONSHIP: [the human 'Charlie Chaplin'] — Tangible concrete thing
Relationship sustainable: **PASS**
Reasoning: Human is Tangible concrete; so is this instance. PASS.

RELATIONSHIP: [the human 'Charlie Chaplin'] — Thing
Relationship sustainable: PASS
Reasoning: 56-33: concrete thing → Thing. PASS.

RELATIONSHIP: [the human 'Charlie Chaplin'] — Composite thing
Relationship sustainable: PASS
Reasoning: Human body is Composite (66-25). PASS.

RELATIONSHIP: [the human 'Charlie Chaplin'] — Composite tangible concrete thing
Relationship sustainable: PASS
Reasoning: PASS.

RELATIONSHIP: (the human 'Charlie Chaplin') — in a mind
Relationship sustainable: PASS
Reasoning: 56-33: Conceptual thing; 59-10: Conceptual things are in the mind. PASS.

RELATIONSHIP: (the human 'Charlie Chaplin') — can be represented by a Conceptual thing
Relationship sustainable: PASS
Reasoning: The discourse never restricts representation to Concrete referents only (56-21 gives the primary case). A Conceptual thing can itself be the referent of another thought. PASS.

RELATIONSHIP: (the human 'Charlie Chaplin') — collection of Things
Relationship sustainable: PASS
Reasoning: Conceptual mirror of a Composite Concrete thing; mirrors its component structure. 56-28: Conceptual things have assigned Features corresponding to the Concrete thing's components. PASS.

RELATIONSHIP: (the human 'Charlie Chaplin') — distinguishable
Relationship sustainable: PASS
Reasoning: 56-28: Conceptual things have Features; 58-08: distinguishable through Features. PASS.

RELATIONSHIP: (the human 'Charlie Chaplin') — with Object
Relationship sustainable: PASS
Reasoning: 56-28: Conceptual things are Conceptual objects with Features. PASS.

RELATIONSHIP: (the human 'Charlie Chaplin') — with assigned Features
Relationship sustainable: PASS
Reasoning: 56-28. PASS.

RELATIONSHIP: (the human 'Charlie Chaplin') — Composite thing
Relationship sustainable: PASS
Reasoning: Conceptual mirror of Composite Concrete thing. 65-02 places it in a relational context indicating multiple relational features. PASS.

RELATIONSHIP: (the human 'Charlie Chaplin') — Noetic thing
Relationship sustainable: PASS
Reasoning: Noetic is placeholder superconcept for all things in the mind. Conceptual things are in the mind (59-10), hence Noetic. PASS.

RELATIONSHIP: (the human 'Charlie Chaplin') — Composite conceptual thing
Relationship sustainable: PASS
Reasoning: Composite + Conceptual. PASS.

RELATIONSHIP: (the human 'Charlie Chaplin') — Composite noetic thing
Relationship sustainable: PASS

Reasoning: Composite + Noetic. PASS.

RELATIONSHIP: <a date> — in a mind
Relationship sustainable: **PASS**
Reasoning: Same as <a number>: Abstract things are conceived in the mind (58-04). PASS.

RELATIONSHIP: <a date> — can be represented by a Conceptual thing
Relationship sustainable: **PASS**
Reasoning: Same as <a number>: Abstract things are representable by Conceptual things. PASS.

RELATIONSHIP: <a date> — single Thing
Relationship sustainable: **PASS**
Reasoning: A specific date is a single Abstract thing. PASS.

RELATIONSHIP: <a date> — distinguishable
Relationship sustainable: **PASS**
Reasoning: 58-08. PASS.

RELATIONSHIP: <a date> — with Object
Relationship sustainable: **PASS**
Reasoning: 56-20 + 58-04 analogy. PASS.

RELATIONSHIP: <a date> — with assigned Features
Relationship sustainable: **PASS**
Reasoning: 56-02 + 58-04. PASS.

RELATIONSHIP: <a date> — cannot represent a Thing
Relationship sustainable: **PASS**
Reasoning: 58-03. PASS.

RELATIONSHIP: <a date> — Elemental thing
Relationship sustainable: **PASS**
Reasoning: A specific date is an atomically simple Abstract value. PASS.

RELATIONSHIP: <a date> — Noetic thing
Relationship sustainable: **PASS**
Reasoning: Abstract thing; Noetic is placeholder superconcept. PASS.

RELATIONSHIP: <a date> — Elemental abstract thing
Relationship sustainable: **PASS**
Reasoning: PASS.

RELATIONSHIP: <a date> — Elemental noetic thing
Relationship sustainable: **PASS**
Reasoning: PASS.

RELATIONSHIP: [an insurance policy] — can be represented by a Conceptual thing
Relationship sustainable: **PASS**
Reasoning: 56-21. PASS.

RELATIONSHIP: [an insurance policy] — collection of Things
Relationship sustainable: **PASS**
Reasoning: Insurance policy is Composite; Composite = collection of component Things. PASS.

RELATIONSHIP: [an insurance policy] — distinguishable
Relationship sustainable: PASS
Reasoning: 56-01. PASS.

RELATIONSHIP: [an insurance policy] — with Object
Relationship sustainable: PASS
Reasoning: 66-08. PASS.

RELATIONSHIP: [an insurance policy] — with assigned Features
Relationship sustainable: PASS
Reasoning: 56-02. PASS.

RELATIONSHIP: [an insurance policy] — conceived
Relationship sustainable: PASS
Reasoning: 56-29: intangible concrete things exist in a conceived world. PASS.

RELATIONSHIP: [an insurance policy] — Intangible concrete thing
Relationship sustainable: PASS
Reasoning: 59-08: insurance policy is intangible (a contract). 59-09: intangible concrete things. PASS.

RELATIONSHIP: [an insurance policy] — Composite thing
Relationship sustainable: PASS
Reasoning: Insurance policy has component clauses, parties, coverage terms, dates. Nothing asserts it Elemental. PASS.

RELATIONSHIP: [an insurance policy] — Composite intangible concrete thing
Relationship sustainable: PASS
Reasoning: PASS.

RELATIONSHIP: [an employment contract] — can be represented by a Conceptual thing
Relationship sustainable: PASS
Reasoning: 56-21. PASS.

RELATIONSHIP: [an employment contract] — collection of Things
Relationship sustainable: PASS
Reasoning: Employment contract is Composite; Composite = collection of component Things. PASS.

RELATIONSHIP: [an employment contract] — distinguishable
Relationship sustainable: PASS
Reasoning: 56-01. PASS.

RELATIONSHIP: [an employment contract] — with Object
Relationship sustainable: PASS
Reasoning: 66-08. PASS.

RELATIONSHIP: [an employment contract] — with assigned Features
Relationship sustainable: PASS
Reasoning: 56-02. PASS.

RELATIONSHIP: [an employment contract] — conceived
Relationship sustainable: PASS
Reasoning: 56-29. PASS.

RELATIONSHIP: [an employment contract] — Intangible concrete thing
Relationship sustainable: PASS
Reasoning: 59-08 + 59-09. PASS.

RELATIONSHIP: [an employment contract] — Composite thing
Relationship sustainable: PASS
Reasoning: Has component aspects (parties, role, salary, dates, terms). Nothing asserts it Elemental. PASS.

RELATIONSHIP: [an employment contract] — Composite intangible concrete thing
Relationship sustainable: PASS
Reasoning: PASS.

RELATIONSHIP: [a car] — can be represented by a Conceptual thing
Relationship sustainable: PASS
Reasoning: 56-21. PASS.

RELATIONSHIP: [a car] — distinguishable
Relationship sustainable: PASS
Reasoning: 56-01. PASS.

RELATIONSHIP: [a car] — with Object
Relationship sustainable: PASS
Reasoning: 66-08. PASS.

RELATIONSHIP: [a car] — with assigned Features
Relationship sustainable: PASS
Reasoning: 56-02. PASS.

RELATIONSHIP: [a car] — physical dimensions
Relationship sustainable: PASS
Reasoning: A car is a tangible physical object. PASS.

RELATIONSHIP: [a car] — Tangible concrete thing
Relationship sustainable: PASS
Reasoning: A car is physically observable — tangible. PASS.

RELATIONSHIP: [a car] — Composite tangible concrete thing
Relationship sustainable: PASS
Reasoning: Car is Composite, Tangible, and Concrete. PASS.

RELATIONSHIP: [a house] — can be represented by a Conceptual thing
Relationship sustainable: PASS
Reasoning: 56-21. PASS.

RELATIONSHIP: [a house] — collection of Things
Relationship sustainable: PASS
Reasoning: House is Composite (rooms, walls, roof, foundation); Composite = collection of component Things. PASS.

RELATIONSHIP: [a house] — distinguishable
Relationship sustainable: PASS
Reasoning: 56-01. PASS.

RELATIONSHIP: [a house] — with Object
Relationship sustainable: PASS

Reasoning: 66-08. PASS.

RELATIONSHIP: [a house] — with assigned Features
Relationship sustainable: **PASS**
Reasoning: 56-02. PASS.

RELATIONSHIP: [a house] — physical dimensions
Relationship sustainable: **PASS**
Reasoning: A house is a physical structure — tangible. PASS.

RELATIONSHIP: [a house] — Composite thing
Relationship sustainable: **PASS**
Reasoning: A house consists of rooms, walls, floors, roof, foundation. Nothing asserts it Elemental. PASS.

RELATIONSHIP: [a house] — Composite tangible concrete thing
Relationship sustainable: **PASS**
Reasoning: PASS.

RELATIONSHIP: [a student] — in reality
Relationship sustainable: **PASS**
Reasoning: Students are humans (Concrete things, 59-08) in reality. PASS.

RELATIONSHIP: [a student] — can be represented by a Conceptual thing
Relationship sustainable: **PASS**
Reasoning: 56-21. PASS.

RELATIONSHIP: [a student] — collection of Things
Relationship sustainable: **PASS**
Reasoning: Student is a human — Composite; Composite = collection of component Things. PASS.

RELATIONSHIP: [a student] — distinguishable
Relationship sustainable: **PASS**
Reasoning: 56-01. PASS.

RELATIONSHIP: [a student] — with Object
Relationship sustainable: **PASS**
Reasoning: 66-08. PASS.

RELATIONSHIP: [a student] — with assigned Features
Relationship sustainable: **PASS**
Reasoning: 56-02. PASS.

RELATIONSHIP: [a student] — physical dimensions
Relationship sustainable: **PASS**
Reasoning: A student is a human — tangible. PASS.

RELATIONSHIP: [a student] — Tangible concrete thing
Relationship sustainable: **PASS**
Reasoning: A student is a human — tangible concrete. PASS.

RELATIONSHIP: [a student] — Thing
Relationship sustainable: **PASS**
Reasoning: Student is a Concrete thing → Thing. PASS.

RELATIONSHIP: [a student] — Composite thing
Relationship sustainable: **PASS**
Reasoning: A student is a human — Composite (66-25). PASS.

RELATIONSHIP: [a student] — Composite tangible concrete thing
Relationship sustainable: **PASS**
Reasoning: PASS.

RELATIONSHIP: (a chair) — in a mind
Relationship sustainable: **PASS**
Reasoning: 65-02 uses parenthesis notation = Conceptual; 59-10: Conceptual things are in the mind. PASS.

RELATIONSHIP: (a chair) — can represent a Thing
Relationship sustainable: **PASS**
Reasoning: 56-21: Conceptual things represent Concrete Things. (a chair) is Conceptual, so it represents [a chair]. PASS.

RELATIONSHIP: (a chair) — can be represented by a Conceptual thing
Relationship sustainable: **PASS**
Reasoning: The discourse never restricts representation to Concrete referents only. A Conceptual thing can itself be represented by another thought. PASS.

RELATIONSHIP: (a chair) — collection of Things
Relationship sustainable: **PASS**
Reasoning: (a chair) is Conceptual mirror of a physical chair (Composite: seat, legs, back). PASS.

RELATIONSHIP: (a chair) — distinguishable
Relationship sustainable: **PASS**
Reasoning: 56-28: Conceptual things have Features; 58-08. PASS.

RELATIONSHIP: (a chair) — with Object
Relationship sustainable: **PASS**
Reasoning: 56-28: Conceptual things are objects with Features. PASS.

RELATIONSHIP: (a chair) — with assigned Features
Relationship sustainable: **PASS**
Reasoning: 56-28. PASS.

RELATIONSHIP: (a chair) — Conceptual thing
Relationship sustainable: **PASS**
Reasoning: 65-02 parenthesis notation; Constraint 8 bracket convention. PASS.

RELATIONSHIP: (a chair) — Composite thing
Relationship sustainable: **PASS**
Reasoning: Conceptual mirror of a physical chair (seat, legs, back). Nothing asserts it Elemental. PASS.

RELATIONSHIP: (a chair) — Noetic thing
Relationship sustainable: **PASS**
Reasoning: Conceptual things are in the mind (59-10); Noetic is placeholder superconcept for all things in the mind. PASS.

RELATIONSHIP: (a chair) — Composite conceptual thing
Relationship sustainable: **PASS**
Reasoning: Composite + Conceptual. PASS.

RELATIONSHIP: (a chair) — Composite noetic thing
Relationship sustainable: **PASS**
Reasoning: Composite + Noetic. PASS.

RELATIONSHIP: [a ship] — in reality
Relationship sustainable: **PASS**
Reasoning: 66-19: concrete ships. PASS.

RELATIONSHIP: [a ship] — can be represented by a Conceptual thing
Relationship sustainable: **PASS**
Reasoning: 56-21. PASS.

RELATIONSHIP: [a ship] — distinguishable
Relationship sustainable: **PASS**
Reasoning: 56-01. PASS.

RELATIONSHIP: [a ship] — physical dimensions
Relationship sustainable: **PASS**
Reasoning: A ship is a physical vessel — tangible. PASS.

RELATIONSHIP: [a ship] — Tangible concrete thing
Relationship sustainable: **PASS**
Reasoning: A ship is a physical, observable structure. PASS.

RELATIONSHIP: [a ship] — Composite tangible concrete thing
Relationship sustainable: **PASS**
Reasoning: PASS.

RELATIONSHIP: [a plank] — in reality
Relationship sustainable: **PASS**
Reasoning: 66-05 context: planks are physical parts of a ship. PASS.

RELATIONSHIP: [a plank] — can be represented by a Conceptual thing
Relationship sustainable: **PASS**
Reasoning: 56-21. PASS.

RELATIONSHIP: [a plank] — collection of Things
Relationship sustainable: **PASS**
Reasoning: A plank is a physical object composed of wood fibres, grain, etc. Composite = collection of component Things. PASS.

RELATIONSHIP: [a plank] — distinguishable
Relationship sustainable: **PASS**
Reasoning: 56-01. PASS.

RELATIONSHIP: [a plank] — with Object
Relationship sustainable: **PASS**
Reasoning: 66-08. PASS.

RELATIONSHIP: [a plank] — with assigned Features
Relationship sustainable: **PASS**
Reasoning: 56-02. PASS.

RELATIONSHIP: [a plank] — physical dimensions

Relationship sustainable: PASS
Reasoning: A plank has physical dimensions. PASS.

RELATIONSHIP: [a plank] — Concrete thing
Relationship sustainable: PASS
Reasoning: 66-05 context; plank is a physical Concrete thing. PASS.

RELATIONSHIP: [a plank] — Tangible concrete thing
Relationship sustainable: PASS
Reasoning: A plank is physically observable. PASS.

RELATIONSHIP: [a plank] — Composite thing
Relationship sustainable: PASS
Reasoning: A plank is composed of wood fibres. Nothing asserts it Elemental. PASS.

RELATIONSHIP: [a plank] — Comcposite concrete thing
Relationship sustainable: PASS
Reasoning: PASS.

RELATIONSHIP: [a plank] — Composite tangible concrete thing
Relationship sustainable: PASS
Reasoning: PASS.

RELATIONSHIP: [a butterfly] — in reality
Relationship sustainable: PASS
Reasoning: Concrete thing — in reality. PASS.

RELATIONSHIP: [a butterfly] — can be represented by a Conceptual thing
Relationship sustainable: PASS
Reasoning: 56-21. PASS.

RELATIONSHIP: [a butterfly] — collection of Things
Relationship sustainable: PASS
Reasoning: Butterfly is a biological organism with component parts. PASS.

RELATIONSHIP: [a butterfly] — distinguishable
Relationship sustainable: PASS
Reasoning: 56-01. PASS.

RELATIONSHIP: [a butterfly] — with assigned Features
Relationship sustainable: PASS
Reasoning: 56-02. PASS.

RELATIONSHIP: [a butterfly] — physical dimensions
Relationship sustainable: PASS
Reasoning: Tangible organism. PASS.

RELATIONSHIP: [a butterfly] — Concrete thing
Relationship sustainable: PASS
Reasoning: 61-06: 'a new entity of the type butterfly' — Entity = Concrete thing. PASS.

RELATIONSHIP: [a butterfly] — Tangible concrete thing
Relationship sustainable: PASS

Reasoning: Butterfly is physically observable. PASS.

RELATIONSHIP: [a butterfly] — Composite thing
Relationship sustainable: **PASS**
Reasoning: Biological organism with component parts. Nothing asserts it Elemental. PASS.

RELATIONSHIP: [a butterfly] — Comcposite concrete thing
Relationship sustainable: **PASS**
Reasoning: PASS.

RELATIONSHIP: [a butterfly] — Composite tangible concrete thing
Relationship sustainable: **PASS**
Reasoning: PASS.

RELATIONSHIP: [a caterpillar] — in reality
Relationship sustainable: **PASS**
Reasoning: Concrete thing — in reality. PASS.

RELATIONSHIP: [a caterpillar] — can be represented by a Conceptual thing
Relationship sustainable: **PASS**
Reasoning: 56-21. PASS.

RELATIONSHIP: [a caterpillar] — collection of Things
Relationship sustainable: **PASS**
Reasoning: Biological organism with component parts. PASS.

RELATIONSHIP: [a caterpillar] — distinguishable
Relationship sustainable: **PASS**
Reasoning: 56-01. PASS.

RELATIONSHIP: [a caterpillar] — with assigned Features
Relationship sustainable: **PASS**
Reasoning: 56-02. PASS.

RELATIONSHIP: [a caterpillar] — physical dimensions
Relationship sustainable: **PASS**
Reasoning: Tangible organism. PASS.

RELATIONSHIP: [a caterpillar] — Concrete thing
Relationship sustainable: **PASS**
Reasoning: 61-12: 'entity of type caterpillar' — Entity = Concrete thing. PASS.

RELATIONSHIP: [a caterpillar] — Tangible concrete thing
Relationship sustainable: **PASS**
Reasoning: PASS.

RELATIONSHIP: [a caterpillar] — Composite thing
Relationship sustainable: **PASS**
Reasoning: Biological organism. Nothing asserts it Elemental. PASS.

RELATIONSHIP: [a caterpillar] — Comcposite concrete thing
Relationship sustainable: **PASS**
Reasoning: PASS.

RELATIONSHIP: [a caterpillar] — Composite tangible concrete thing
Relationship sustainable: PASS
Reasoning: PASS.

RELATIONSHIP: [a brain] — in reality
Relationship sustainable: PASS
Reasoning: Physical organ — in reality. PASS.

RELATIONSHIP: [a brain] — can be represented by a Conceptual thing
Relationship sustainable: PASS
Reasoning: 56-21. PASS.

RELATIONSHIP: [a brain] — collection of Things
Relationship sustainable: PASS
Reasoning: Brain is a biological organ with neurons, lobes, etc. Composite = collection of component Things. PASS.

RELATIONSHIP: [a brain] — distinguishable
Relationship sustainable: PASS
Reasoning: 56-01. PASS.

RELATIONSHIP: [a brain] — with Object
Relationship sustainable: PASS
Reasoning: 66-08. PASS.

RELATIONSHIP: [a brain] — with assigned Features
Relationship sustainable: PASS
Reasoning: 56-02. PASS.

RELATIONSHIP: [a brain] — physical dimensions
Relationship sustainable: PASS
Reasoning: Physical organ — tangible. PASS.

RELATIONSHIP: [a brain] — Concrete thing
Relationship sustainable: PASS
Reasoning: Physical organ — Concrete thing. PASS.

RELATIONSHIP: [a brain] — Tangible concrete thing
Relationship sustainable: PASS
Reasoning: PASS.

RELATIONSHIP: [a brain] — Composite thing
Relationship sustainable: PASS
Reasoning: Brain is a biological organ with neurons, lobes, etc. Nothing asserts it Elemental. PASS.

RELATIONSHIP: [a brain] — Comcposite concrete thing
Relationship sustainable: PASS
Reasoning: PASS.

RELATIONSHIP: [a brain] — Composite tangible concrete thing
Relationship sustainable: PASS
Reasoning: PASS.

RELATIONSHIP: [a motor block] — in reality
Relationship sustainable: PASS
Reasoning: Physical engine part — in reality. PASS.

RELATIONSHIP: [a motor block] — can be represented by a Conceptual thing
Relationship sustainable: PASS
Reasoning: 56-21. PASS.

RELATIONSHIP: [a motor block] — collection of Things
Relationship sustainable: PASS
Reasoning: Motor block has cylinders, pistons, valves. Composite = collection of component Things. PASS.

RELATIONSHIP: [a motor block] — distinguishable
Relationship sustainable: PASS
Reasoning: 56-01. PASS.

RELATIONSHIP: [a motor block] — with Object
Relationship sustainable: PASS
Reasoning: 66-08. PASS.

RELATIONSHIP: [a motor block] — with assigned Features
Relationship sustainable: PASS
Reasoning: 56-02. PASS.

RELATIONSHIP: [a motor block] — physical dimensions
Relationship sustainable: PASS
Reasoning: Physical engine part — tangible. PASS.

RELATIONSHIP: [a motor block] — Concrete thing
Relationship sustainable: PASS
Reasoning: Physical engine part — Concrete thing. PASS.

RELATIONSHIP: [a motor block] — Tangible concrete thing
Relationship sustainable: PASS
Reasoning: PASS.

RELATIONSHIP: [a motor block] — Composite thing
Relationship sustainable: PASS
Reasoning: Motor block has cylinders, pistons, valves. Nothing asserts it Elemental. PASS.

RELATIONSHIP: [a motor block] — Comcposite concrete thing
Relationship sustainable: PASS
Reasoning: PASS.

RELATIONSHIP: [a motor block] — Composite tangible concrete thing
Relationship sustainable: PASS
Reasoning: PASS.

---

6. Check 5 — Unmarked Relationships (– –)

Objective: Verify that selected '– –' cells cannot be derived from the discourse or formal context — i.e., their absence is correct.

Result: All 20 tested absent cells are CORRECTLY ABSENT.

ABSENT: [a cloud] — in a mind
Correctly absent: PASS
Reasoning: Cloud is Concrete (in reality). By disjoint & complete 'in a mind / in reality' constraint: correctly absent.

ABSENT: [a cloud] — can represent a Thing
Correctly absent: PASS
Reasoning: Only Conceptual things represent other things (56-21). Cloud is Concrete, not Conceptual. Correctly absent.

ABSENT: [a cloud] — single Thing
Correctly absent: PASS
Reasoning: Cloud is Composite (56-05: water drops and crystals). By disjoint & complete Composite/Elemental partition: correctly absent.

ABSENT: [a cloud] — cannot represent a Thing
Correctly absent: PASS
Reasoning: 58-03 applies to Abstract things. Cloud is Concrete, not Abstract. Correctly absent.

ABSENT: [a cloud] — Abstract thing
Correctly absent: PASS
Reasoning: Cloud is Concrete. Disjoint Concrete/Abstract constraint. Correctly absent.

ABSENT: [a cloud] — conceived
Correctly absent: PASS
Reasoning: Cloud is Tangible. Disjoint & complete Tangible/Intangible partition. Correctly absent.

ABSENT: [a cloud] — Intangible concrete thing
Correctly absent: PASS
Reasoning: Cloud is Tangible. Disjoint & complete Tangible/Intangible partition. Correctly absent.

ABSENT: [a cloud] — Conceptual thing
Correctly absent: PASS
Reasoning: Cloud is Concrete. Disjoint Concrete/Conceptual constraint. Correctly absent.

ABSENT: [a human] — in a mind
Correctly absent: PASS
Reasoning: Humans are in reality (59-08). Disjoint & complete 'in a mind / in reality'. Correctly absent.

ABSENT: [a human] — cannot represent a Thing
Correctly absent: PASS
Reasoning: 58-03 applies to Abstract things. Humans are Concrete. Correctly absent.

ABSENT: [a human] — Abstract thing
Correctly absent: PASS
Reasoning: Humans are Concrete. Disjoint Concrete/Abstract. Correctly absent.

ABSENT: [a human] — conceived
Correctly absent: PASS
Reasoning: Humans are Tangible (56-29). Disjoint Tangible/Intangible. Correctly absent.

ABSENT: [a human] — Conceptual thing
Correctly absent: PASS

Reasoning: Humans are Concrete. Disjoint Concrete/Conceptual. Correctly absent.

ABSENT: <a number> — can represent a Thing

Correctly absent: PASS

Reasoning: 58-03: Abstract things don't represent Concrete things. Numbers are Abstract. Correctly absent.

ABSENT: <a number> — in reality

Correctly absent: PASS

Reasoning: 58-07: 387 cannot be observed — it is in the mind, not in reality. Disjoint 'in a mind / in reality'. Correctly absent.

ABSENT: <a number> — Concrete thing

Correctly absent: PASS

Reasoning: Numbers are Abstract. Disjoint Concrete/Abstract. Correctly absent.

ABSENT: <a number> — Conceptual thing

Correctly absent: PASS

Reasoning: Abstract things and Conceptual things are both Noetic but are distinct subconcepts. PASS as absent.

ABSENT: (the human 'Charlie Chaplin') — in reality

Correctly absent: PASS

Reasoning: Conceptual things are in the mind (59-10). Disjoint 'in a mind / in reality'. Correctly absent.

ABSENT: (the human 'Charlie Chaplin') — Concrete thing

Correctly absent: PASS

Reasoning: Conceptual and Concrete are disjoint. Correctly absent.

ABSENT: (the human 'Charlie Chaplin') — cannot represent a Thing

Correctly absent: PASS

Reasoning: 58-03 applies to Abstract things. Conceptual things CAN represent things (56-21). Correctly absent.

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7. Check 6 — Sentence Interpretations

Objective: Verify that each interpretation in Thing_D_Discourse.pdf correctly applies the formal context while adhering to the intent of the original sentence.

Result: All 20 sentence interpretations are CORRECT.

INTERPRETATION: 54-01

Original: Thoughts are things in the mind.

Interpretation: Some thoughts are Conceptual things in the mind.

Modification: PASS

Reasoning: Interpretation adds 'Conceptual' qualifier and 'Some'. Conceptual things are things in the mind (59-10); not all thoughts are Conceptual (Abstract thoughts also exist, 58-04). Correct application of EO taxonomy.

INTERPRETATION: 54-02

Original: They represent concrete things (i.e. things outside the mind), called referents.

Interpretation: They represent Concrete things (i.e. Things in reality), called referents in the semiotic triangle.

Modification: PASS

Reasoning: 'Things in reality' is the correct EO equivalent of 'things outside the mind'. Capitalisation consistent with formal context. CORRECT.

INTERPRETATION: 55-12

Original: Bunge defines a world as a collection of things, 'untouched' yet by mental abstractions.

Interpretation: Bunge defines a World as a collection of Things, 'untouched' yet by mental abstractions.

Modification: **PASS**

Reasoning: Capitalisation of World and Things consistent with EO formal context. CORRECT.

INTERPRETATION: 56-01

Original: The basic assumption in world ontology is that a world consists of distinguishable concrete things or things for short.

Interpretation: The basic assumption in world ontology is that a World consists of distinguishable Things.

Modification: **PASS**

Reasoning: Drops 'concrete' and uses 'Things' as the formal context superclass. Valid abstraction: the formal context uses Thing as the top-level term. CORRECT.

INTERPRETATION: 56-02

Original: A Thing is said to be an Object with assigned Features.

Interpretation: A Thing is said to be an Object with an assigned Form of Features.

Modification: **PASS**

Reasoning: Introduces 'Form' as the collection of Features (60-06: 'the Form of a Thing is the collection of its Features'). CORRECT.

INTERPRETATION: 56-13

Original: The key to Bunge's solution of the problem of sameness and change, which we will adopt in Sect. 5.3.2, is that the Features of a Thing are assigned to an Object. Bunge names an Object a bare individual.

Interpretation: The key to Bunge's solution of the problem of sameness and change, which we will adopt in Sect. 5.3.2, is that a Thing is a Form, a collection of Features, that is assigned to an Object. Bunge names an Object a bare individual.

Modification: **PASS**

Reasoning: Reframes 'Features assigned to Object' as 'Thing is a Form assigned to Object'. Consistent with 60-06 and 66-08. CORRECT.

INTERPRETATION: 56-21

Original: When observing and distinguishing the Concrete things in a world, the mind creates Conceptual things that represent the Concrete Things.

Interpretation: When recognising and distinguishing the Things in a world, the mind creates Conceptual things that represent the Concrete Things.

Modification: **PASS**

Reasoning: 'Recognising' replaces 'observing' — consistent with 56-03 ('recognised'). Drops 'Concrete' before 'things in a world' — the formal context uses Thing as superclass. CORRECT.

INTERPRETATION: 56-24

Original: It is called classifying: we observe similarities and differences between things and accordingly put them into different classes.

Interpretation: It is called classifying: we recognise similarities and differences between Things and accordingly put them into different Classes.

Modification: **PASS**

Reasoning: 'Recognise' for 'observe' (56-03); 'Classes' capitalised per formal context. CORRECT.

INTERPRETATION: 56-33

Original: To avoid confusion, we will refer to the concrete thing as <the human 'Charlie Chaplin'>, and to the corresponding conceptual thing as (the human 'Charlie Chaplin').

Interpretation: To avoid confusion, we will refer to the Concrete thing as [the human 'Charlie Chaplin'], and to the corresponding Conceptual thing as (the human 'Charlie Chaplin').

Modification: **PASS**

Reasoning: Standardises to the formal context bracket convention (Constraint 8): [ ] for Concrete, ( ) for Conceptual. The original used < > inconsistently; the interpretation applies the agreed convention. CORRECT.

INTERPRETATION: 57-03

Original: It is represented in our mind by the thought (the class HUMAN).

Interpretation: The Concrete things of Class [HUMAN] are represented by Conceptual things from Class (HUMAN).

Modification: PASS

Reasoning: Applies the formal context's Concrete class [HUMAN] / Conceptual class (HUMAN) distinction (57-02, 60-25). More precise than original but fully consistent with surrounding sentences. CORRECT.

INTERPRETATION: 59-01

Original: If the thing of a Unary fact is a Concrete thing, the Concrete thing is commonly called an Entity.

Interpretation: If the Thing of a Unary fact is a Concrete thing, the Concrete unary fact is commonly called an Entity.

Modification: PASS

Reasoning: Shifts emphasis to 'Concrete unary fact' rather than 'Concrete thing' as what is called Entity. Consistent with the formal context's treatment of Entity as a kind of Fact. CORRECT.

INTERPRETATION: 59-04

Original: As an example, day 2411108 is a value on the time scale Julian Date.

Interpretation: As an example, date 2411108 is a value on the time scale Julian Date.

Modification: PASS

Reasoning: 'day' to 'date' — a date is more precise and consistent with 'Julian Date' as a time scale. CORRECT.

INTERPRETATION: 60-08

Original: In applying a type, the mind puts, so to speak, the type's template over the things in the world, and sieves out the ones that conform.

Interpretation: In applying a type, the mind puts, so to speak, the type's template over the Things in the World, and sieves out the ones that conform.

Modification: PASS

Reasoning: Capitalisation of Things and World consistent with formal context. CORRECT.

INTERPRETATION: 60-21

Original: Let us call the Conceptual class of Things that are predicated by Facts of the Type human, HUMAN.

Interpretation: Let us call the Conceptual class of Conceptual things that are predicated by Facts of the Type human, (HUMAN).

Modification: PASS

Reasoning: Uses ( ) for Conceptual class per Constraint 8 bracket convention. Specifies 'Conceptual things' for clarity. CORRECT.

INTERPRETATION: 60-25

Original: Lastly, the conceptual class (HUMAN) represents the concrete class [HUMAN].

Interpretation: Lastly, the Conceptual things of the Conceptual class (HUMAN) represents the Concrete things of Concrete class [HUMAN].

Modification: PASS

Reasoning: More precise: Conceptual things represent Concrete things (56-21), not the class as an abstract entity represents another class. CORRECT.

INTERPRETATION: 61-04

Original: A Fact starts to exist either as soon as the Form of a thing conforms to a Type, or as soon as a Type starts to exist to which the Thing conforms.

Interpretation: A Fact starts to exist either as soon as the Thing conforms to a Type, or as soon as a Type starts to exist to which the Thing conforms.

Modification: PASS

Reasoning: Drops 'Form of a' — valid simplification since a Thing's conformance to a Type is assessed through its Form. CORRECT.

INTERPRETATION: 61-09

Original: A Fact ends to exist either as soon as the Form of the Thing that the Fact is about, does not conform anymore to the Type, or as soon as a Type to which it conforms, ends to exist.

Interpretation: A Fact ends to exist either as soon as the Thing that the Fact is about, does not conform anymore to the Type, or as soon as a Type to which it conforms, ends to exist.

Modification: PASS

Reasoning: Same simplification as 61-04. CORRECT.

INTERPRETATION: 64-01

Original: The individuals in all these classes are identified by the concrete identity they have got in the class genus.

Interpretation: The Things in all these Classes are identified by the Things in Class genus.

Modification: **PASS**

Reasoning: 'Individuals' → 'Things'; 'classes' → 'Classes'; 'class genus' → 'Class genus'. Applies formal context terminology consistently. CORRECT.

INTERPRETATION: 64-11

Original: The type human, as discussed, is typically a constructional type: it is based exclusively on the properties of things.

Interpretation: The Type human, as discussed, is typically a constructional Type: it is based exclusively on the Form of a Thing.

Modification: **PASS**

Reasoning: 'properties of things' → 'Form of a Thing'. In EO, properties = Features = Form (60-06). CORRECT.

INTERPRETATION: 66-14

Original: This would even hold if the Thing would not conform to the Type ship anymore, as a butterfly has kept its Adorned object after having transformed to being a butterfly from being a caterpillar before.

Interpretation: This would even hold if the Thing would not conform to the Type ship anymore, as a butterfly has kept its Object after having transformed to being a butterfly from being a caterpillar before.

Modification: **PASS**

Reasoning: 'Adorned object' → 'Object'. In EO's core terminology the Object is the bare individual (56-13); 'Adorned object' is not a standard term in the formal context. The interpretation uses the correct formal context term. CORRECT.

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8. Overall Verdict

Check	Entries checked	PASS	FAIL
Check 1 — Items	23 (+ 4 skipped)	23	0
Check 2 — Characteristics	18 (+ 15 skipped)	18	0
Check 3 — Explicit relationships	72	72	0
Check 4 — '– #' relationships	227	227	0
Check 5 — Absent relationships	20 (selected)	20	0
Check 6 — Interpretations	20	20	0

Traceability Assessment

FULLY TRACEABLE.

The formal context for 'Thing' is fully traceable to the EO discourse when assessed at the concept level, as required by the validation principles. Every referenced item, every referenced characteristic, every explicit relationship cell, every inherited relationship cell, and every sentence interpretation is traceable to or consistent with the EO discourse.

The four conceptual pillars that underpin full traceability are:

(1) Composite classification: The Composite / Elemental partition in the formal context, combined with the complete absence of any Elemental assertion for non-placeholder items in the discourse, supports Composite classification for all Things with discernible component parts. This covers biological organisms, physical structures, and intangible contracts alike.

(2) Noetic thing as superconcept: The placeholder term 'Noetic thing' is structurally necessary to unify Conceptual things and Abstract things under their shared 'in a mind' characteristic. Both are thoughts in the mind; they differ only in whether they represent a Concrete thing. The placeholder carries no discourse reference by design and is correctly transparent.

(3) Abstract things in the mind: Both Conceptual things and Abstract things are thoughts in the mind. The number 387 cannot be observed in reality — it is conceived mentally. The 'in a mind' characteristic is therefore correctly attributed to Abstract things.

(4) Representability of Abstract things: The discourse's own use of parenthesis notation for (the number '387') demonstrates a Conceptual thing in the mind that represents an Abstract thing. The discourse never restricts the representation relation to Concrete referents only.